AMERY ICE SHELF BOREHOLE MICROCAT AND CTD DATA, SEASON 2000/2001 TO 2004/2005 - DATA PROCESSING AND QUALITY

Mark Rosenberg (data processor)
Amery Ice Shelf borehole drill team (data collectors)
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AMERY ICE SHELF BOREHOLE MICROCAT AND CTD DATA, SEASON 2000/2001 TO 2004/2005 - DATA PROCESSING AND QUALITY

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1. INTRODUCTION

This report summarises the entire AMISOR (Amery Ice Shelf Experiment) borehole microcat and CTD data set, from the initial season 2000/2001 up to season 2004/2005. Details of data processing and data quality are provided, while updates to the data are described. The latest 2004/2005 microcat data, not described in earlier reports, are included. All borehole microcat, CTD and CTD/microcat comparison information from previous reports (Rosenberg et al., 2002, and Rosenberg et al., unpublished) are included here, to provide a single up to date account of the data. The fieldwork for each season is described in Craven et al. (2000, 2001, 2002, 2004).

At the time of writing, two microcat moorings continue to operate, at borehole sites AM02 and AM01 (Table 1). Each mooring consists of three SeaBird SBE37IM inductive modem microcats, hanging suspended from the underneath of the iceshelf, and with the mooring wire frozen into the iceshelf borehole (Figure 1). Data are downloaded each season via an inductive coupling, connected to the top of the mooring wire protruding from the borehole. To date, three years' worth of temperature and salinity time series data have been obtained from each mooring.

Also at the time of writing, a total of 16 CTD profiles have been taken at three borehole sites: AM02, AM01 and AM01b, in three different seasons (Table 2). Each profile was collected using an FSI 3" MicroCTD, serial 1610, with two Niskin bottles and a bottom detection switch attached inline.

<u>Table 1:</u> (on following page) Station details for AMISOR borehole microcats. Note that borehole absolute locations change in time, as the ice shelf is in motion. For the microcats, positive clock errors indicate microcat clocks were fast.
Notes:
*** positions marked with * are estimates from iceshelf velocity calculations, and should be considered mean positions for the relevant summers**
*** mean microcat depths (mean_cat_depth in table):**
for AM02 dep and AM01 dep - instrument depths after mooring deployments
for AM02 1a, 1, 3 and 4 - mean over data time span
for AM01 1a, 1b, 1, 2 and 3 - mean over first couple of days of deployment in Jan. 2002
*** magnetic declinations are averages over data time spans (except for point values at initial mooring deployments)**
*** bottom depth (total water depth from water surface in borehole to ocean bottom)**
= 790 m at deployment time for AM02
= 783 m at deployment time for AM01
*** Clock errors only indicated for time spans where clock drift correction has been applied. Positive clock errors indicate microcat clocks were fast.**

mooring	*microcat*	*download number,time (UTC)*	*GPS reading for mooring site on given UTC date*	*data time span (UTC)*	*mean_cat_depth depth (m)*	*pressure (dbar)*	*magdec (deg)*	*d(magdec)/dt (deg/year)*	*clock error (sec)*
AM02	1623	dep, 06/01/2001	69deg 42.826'S, 72deg 38.399'E on 06/01/2001	(deployment)	334.8	338.5	-75.10	-0.14	
AM02	1624	dep, 06/01/2001	69deg 42.826'S, 72deg 38.399'E on 06/01/2001	(deployment)	556.3	562.7	-75.10	-0.14	
AM02	1174	dep, 06/01/2001	69deg 42.826'S, 72deg 38.399'E on 06/01/2001	(deployment)	761.0	770.2	-75.10	-0.14	
AM02	1623	1a, 14/02/2001	-	06/01/2001-14/02/2001	335.0	338.7	-75.10	-0.14	-
AM02	1624	1a, 14/02/2001	-	06/01/2001-14/02/2001	556.5	562.9	-75.10	-0.14	-
AM02	1174	1a, 14/02/2001	-	06/01/2001-14/02/2001	761.9	771.1	-75.10	-0.14	-
AM02	1623	1, 06/02/2002	*69deg 42.564'S, 72deg 39.150'E on 06/02/2002	06/01/2001-06/02/2002	334.9	338.6	-75.18	-0.14	+120
AM02	1624	1, 06/02/2002	*69deg 42.564'S, 72deg 39.150'E on 06/02/2002	06/01/2001-06/02/2002	556.6	563.0	-75.18	-0.14	+300
AM02	1174	1, 06/02/2002	*69deg 42.564'S, 72deg 39.150'E on 06/02/2002	06/01/2001-06/02/2002	762.8	772.0	-75.18	-0.14	+120
AM02	1623	-, 05/02/2003	*69deg 42.322'S, 72deg 39.901'E on 05/02/2003	(logging failed)	-	-	-75.32	-0.14	-
AM02	1624	-, 05/02/2003	*69deg 42.322'S, 72deg 39.901'E on 05/02/2003	(logging failed)	-	-	-75.32	-0.14	-
AM02	1174	-, 05/02/2003	*69deg 42.322'S, 72deg 39.901'E on 05/02/2003	(logging failed)	-	-	-75.32	-0.14	-
AM02	1623	3, 28/12/2003	69deg 42.146'S, 72deg 40.423'E on 28/12/2003	05/02/2003-28/12/2003	334.0	337.7	-75.45	-0.14	-
AM02	1624	3, 28/12/2003	69deg 42.146'S, 72deg 40.423'E on 28/12/2003	05/02/2003-28/12/2003	555.9	562.4	-75.45	-0.14	-
AM02	1174	3, 28/12/2003	69deg 42.146'S, 72deg 40.423'E on 28/12/2003	05/02/2003-28/12/2003	762.9	772.1	-75.45	-0.14	-
AM02	1623	4, 26/12/2004	69deg 41.911'S, 72deg 41.104'E on 30/12/2004	02/01/2004-26/12/2004	333.6	337.2	-75.59	-0.14	+104
AM02	1624	4, 26/12/2004	69deg 41.911'S, 72deg 41.104'E on 30/12/2004	28/12/2003-26/12/2004	555.5	562.0	-75.59	-0.14	+255
AM02	1174	4, 26/12/2004	69deg 41.911'S, 72deg 41.104'E on 30/12/2004	28/12/2003-26/12/2004	762.7	771.9	-75.59	-0.14	+109
AM01	1969	dep, 15/01/2002	69deg 26.522'S, 71deg 25.025'E on 15/01/2002	(deployment)	436.2	441.1	-74.03	-0.14	
AM01	1970	dep, 15/01/2002	69deg 26.522'S, 71deg 25.025'E on 15/01/2002	(deployment)	574.9	581.6	-74.03	-0.14	
AM01	1971	dep, 15/01/2002	69deg 26.522'S, 71deg 25.025'E on 15/01/2002	(deployment)	734.6	743.4	-74.03	-0.14	
AM01	1969	1a, 23/01/2002	-	15/01/2002-23/01/2002	436.2	441.1	-74.03	-0.14	-
AM01	1970	1a, 23/01/2002	-	15/01/2002-23/01/2002	574.8	581.5	-74.03	-0.14	-
AM01	1971	1a, 23/01/2002	-	15/01/2002-23/01/2002	734.2	743.0	-74.03	-0.14	-
AM01	1969	1b, 17/11/2002	*69deg 26.212'S, 71deg 25.875'E on 17/11/2002	23/01/2002-17/11/2002	436.2	441.1	-74.09	-0.14	-
AM01	1970	1b, 17/11/2002	*69deg 26.212'S, 71deg 25.875'E on 17/11/2002	23/01/2002-17/11/2002	574.8	581.5	-74.09	-0.14	-
AM01	1971	1b, 17/11/2002	*69deg 26.212'S, 71deg 25.875'E on 17/11/2002	23/01/2002-17/11/2002	734.2	743.0	-74.09	-0.14	-
AM01	1969	1, 26/01/2003	-	15/01/2002-26/01/2003	436.2	441.1	-74.10	-0.14	-
AM01	1970	1, 26/01/2003	-	15/01/2002-26/01/2003	574.8	581.5	-74.10	-0.14	-
AM01	1971	1, 26/01/2003	-	15/01/2002-26/01/2003	734.2	743.0	-74.10	-0.14	-
AM01	1969	2, 25/12/2003	69deg 25.933'S, 71deg 26.712'E on 25/12/2003	26/01/2003-25/12/2003	436.2	441.1	-74.24	-0.14	-
AM01	1970	2, 26/12/2003	69deg 25.933'S, 71deg 26.712'E on 25/12/2003	26/01/2003-26/12/2003	574.8	581.5	-74.24	-0.14	-
AM01	1971	2, 26/12/2003	69deg 25.933'S, 71deg 26.712'E on 25/12/2003	26/01/2003-26/12/2003	734.2	743.0	-74.24	-0.14	-
AM01	1969	3, 30/12/2004	69deg 25.610'S, 71deg 27.593'E on 04/01/2005	26/12/2003-30/12/2004	436.2	441.1	-74.38	-0.14	+163
AM01	1970	3, 30/12/2004	69deg 25.610'S, 71deg 27.593'E on 04/01/2005	26/12/2003-30/12/2004	574.8	581.5	-74.38	-0.14	+154
AM01	1971	3, 30/12/2004	69deg 25.610'S, 71deg 27.593'E on 04/01/2005	26/12/2003-30/12/2004	734.2	743.0	-74.38	-0.14	+182

2. MICROCAT DATA

2.1 General

Microcat data, logged at a 30 minute recording interval, are downloaded when the borehole locations are visited each season. Data are downloaded to a PC, using the terminal program Seaterm. Ideally, clock errors are noted prior to download, and clocks are reset to GPS UTC time prior to recommencement of logging.

Manufacturer supplied calibrations (July/August/September 2000 for AM02 instruments, May/June 2001 for AM01 instruments) are applied internally by the microcats, thus calibrated data are output. Data are then processed, padding files at start and end, checking for and filling data gaps, calculating decimal times, and checking for bad data. For decimal times, all microcats are assigned the same decimal time scheme, with decimal days starting from midnight on December 31st 2000 (so, for example, decimal time =0.5 for midday on January 1st 2001; =365.5 for midday on January 1st 2002).

For cases where a reliable clock drift error has been obtained (Table 1), times are "compressed" linearly throughout the relevant part of the time series to correct for this error. This may result in a small time discontinuity where one data segment joins another. Time-corrected data have not been re-interpolated onto regular time intervals. For cases where there is no reliable clock drift error measurement (Table 1), the original times (as recorded by the microcats) have been retained. This results in a small time inaccuracy, up to 5 minutes maximum.

Microcat time series are plotted in Figures 2 and 3. A few smaller conductivity/salinity spikes are evident, but they have not been removed from the data. Only larger spikes have been removed, as detailed below. AM02 microcat pressure sensor data are plotted in Figure 4.

2.2 Microcat data processing notes

For AM02, the pressure/depth values showed in Table 1 are mean values calculated from the pressure time series of each data download. For AM01, the pressure sensors were faulty for all 3 microcats, although a small amount of reliable pressure data were recorded for the first few days after initial deployment. Pressure/depth values used for all AM01 microcat data are determined from the first few pressure sensor values after deployment. Salinity and conductivity values for all AM01 microcat data are recalculated using these constant pressure values.

AM02, download 1: For the first few weeks following deployment, communication was made with the 3 microcats on several occasions. After each communication, the "delayed start" feature was not used, and logging thus recommenced on the completion of communication. This occurred at a different part of the hour on each occasion. As a consequence, there are several time discontinuities through the time series from download 1. The discontinuities are at the following times:

microcat 1623 - ~1630 on 09/01/2001; ~0430 on 16/01/2001; ~0630 on 14/02/2001
microcat 1624 - ~1700 on 09/01/2001; ~0500 on 16/01/2001; ~0730 on 14/02/2001
microcat 1174 - ~1700 on 09/01/2001; ~0500 on 16/01/2001; ~0800 on 14/02/2001

AM02, download 3, microcat 1623: A break in data logging occurred from 1100 on 1st October 2003 to 1900 on 4th October 2003 (reason unknown).

AM02, download 4, microcat 1623: Data logging appeared to commence on 2nd January 2004, 5 days after the 2 adjacent microcats; reason unknown.

AM02, download 4, microcat 1174: Good conductivity/salinity data ends on 2nd March 2004. After this time, conductivity took a permanent negative step, of the order ~2 mS/cm. The resulting bad conductivity and salinity data have been removed. The cause of the data step (fouling or electronic fault) is unknown. This is most likely terminal for microcat 1174 salinity.

AM01, download 3: The following data spikes were removed from the data:

microcat 1969 - temperature spike at 0000 on 20/03/2004
microcat 1969 - conductivity spike at 0130 on 02/04/2004
microcat 1969 - temperature spike at 1500 on 18/06/2004

2.3 Microcat data accuracy notes

(a) sensor calibration drift

Calibration drift of the microcat sensors is to be expected after several years in the water. The manufacturer quotes the following "typical stability" for sensors:

0.003 mS/cm per month for conductivity
0.0002^{o}C per month for temperature
0.004% of full scale range per month for pressure

Practical field experience of sensor drift was gained from microcat deployments in and around the Mertz Polynya on the Antarctic continental shelf (Rosenberg et al., 2001), where 18 microcats were deployed for 22 months, and a further 9 microcats were deployed for 6.5 months. Comparison of pre and post deployment calibrations revealed sensor drift, with a mean value of 0.0003 (PSS78) **per month** for salinity, and 0.0005^{o}C **per year** for temperature. Sensors with incremental calibration shifts (due to hardware problems) were excluded from these means. Translating these values for the Mertz Polynya deployments into potential drift for the AMISOR microcat sensors, we might expect the following at the times of the various data downloads:

borehole	download	~time since deployment	total salinity drift (PSS78)	total temperature drift (^{o}C)
AM02	1	1 year	0.004	0.0005
AM02	3	3 years	0.011	0.0015
AM02	4	4 years	0.014	0.0020
AM01	1	1 years	0.004	0.0005
AM01	2	2 years	0.007	0.0010
AM01	3	3 years	0.011	0.0015

(b) salinity error due to pressure

For AM01, constant pressure values (derived from the first few readings after deployment) were used for the recalculation of conductivity and salinity, as discussed previously. As pressure fluctuations for AM01 are unknown, a small salinity error is therefore expected. The magnitude of this error can be estimated from examination of AM02 microcat pressure data (Figure 4). From the figure, pressures fluctuate by up to a maximum of 3 dbar over the time series, with only a few exceptions. For AM01, at the local conditions of conductivity ~27 mS/cm and temperature ~-2°C, omission of pressure fluctuations of up to 3 dbar from salinity calculations would result in a maximum salinity error of ~0.002 (PSS78).

Additional salinity errors for AM01 might arise from unknown mean vertical changes in mooring position over the years. From Figure 4, AM02 microcat pressure data show a decrease in mean pressure, over 4 years, of the order 1 to 2 dbar. However, contributions to this shift could come from both real physical change, and from pressure sensor calibration drift. Whatever the case, the salinity error resulting from a 1 to 2 dbar pressure change is considered negligible here, compared with the errors caused by calibration drift of temperature and conductivity sensors discussed in the previous section.

(c) time inaccuracies

For AM02 download 3, and AM01 downloads 1 and 2, no reliable clock drift errors were obtained. Times in the data files for these periods (Table 1) should only be considered accurate to 5 minutes.

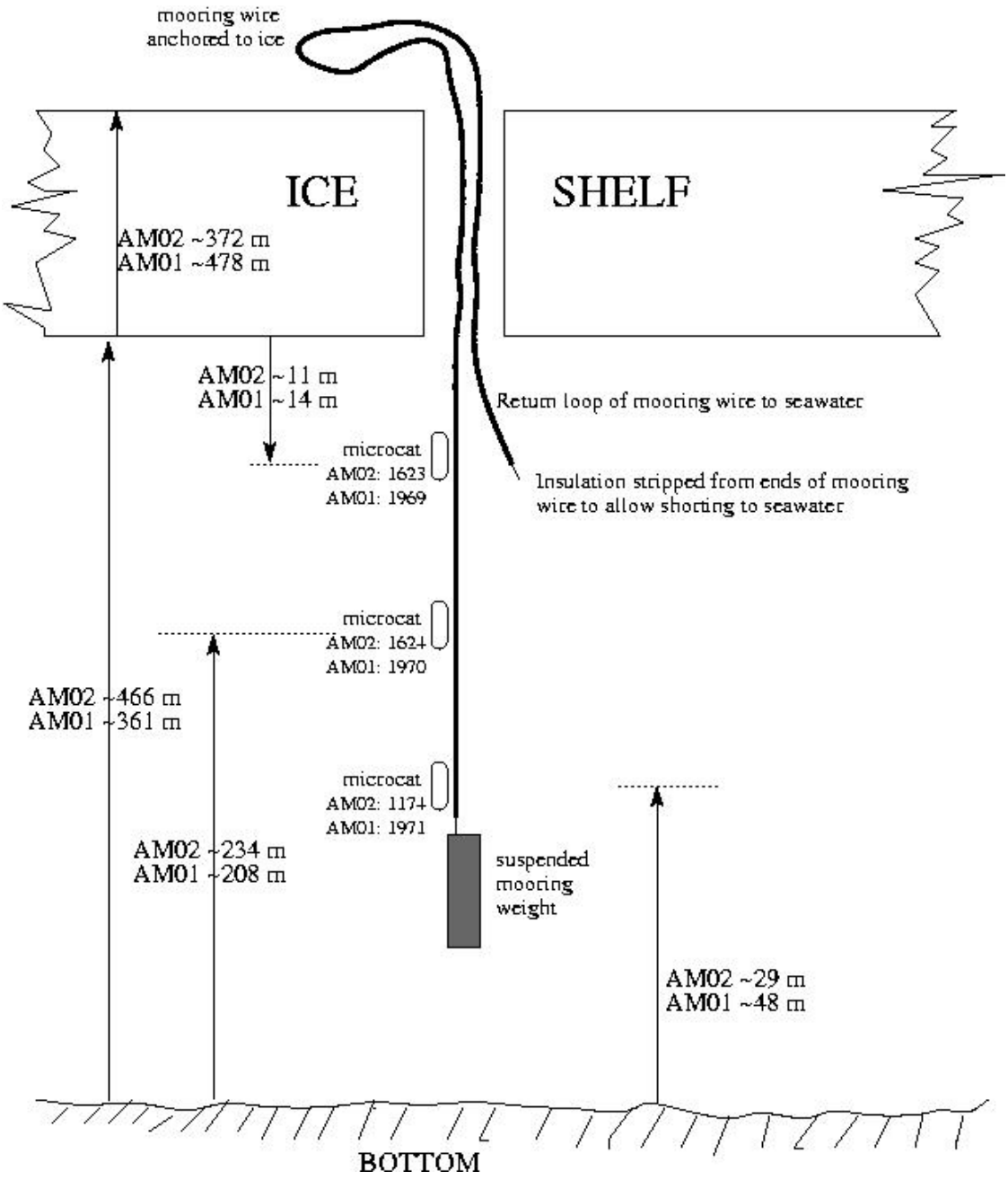
3. BOREHOLE MICROCAT/CTD DATA COMPARISON

AM02 and AM01

A brief comparison was made between borehole microcat and CTD temperature and salinity data, for the CTD data collected at AM02 in the 2000/2001 season (section 4 of this report), and for the CTD data collected at AM01 in the 2001/2002 season (section 4 of this report). Although no simultaneous microcat and CTD measurements exist, the time difference was only of the order of several days, and a valid comparison can still be made in TS space. Fairly good agreement was found between the CTD and microcat data for borehole AM01 (Figure 6) in the 2001/2002 season. For the earlier borehole AM02 in the 2000/2001 season, temperatures agree fairly well, but CTD salinities are on average ~0.03 (PSS78) lower than microcat salinities (Figure 5). Note that for AM02, no borehole Niskin bottle samples were available to correct the CTD data (section 4 of this report). However the correction found for the AM01 CTD salinities from the borehole Niskins was +0.028 (section 4 of this report). This value is very close to the above microcat/CTD salinity difference for AM02. It is therefore assumed that the microcat data are correct, and the AM02 CTD salinity values are low by ~0.03.

AM01-B

A brief comparison was made between simultaneous temperature and salinity measurements from borehole AM01 microcat data and borehole AM01-B CTD data (2003/2004 season, section 4 of this report) (Figure 7). Note that the horizontal separation between the two borehole locations was ~1.6 km. Temperatures compare well at the deepest position (744 dbar). Temperatures differ at the other 2 positions, as follows. The middle position (582 dbar) is in the steepest part of the vertical gradient, hence the horizontal spatial separation between the 2 locations may be significant. The top position (442 dbar) at AM01-B lies close to the top of the cold ice shelf water below the ice/water interface (from the temperature profile in Figure 23). At the same depth at AM01, temperature is evidently warmer (from microcat data). For salinity, a general trend is visible, with CTD salinities lower than microcat values by ~0.03 PSS78. This salinity difference excedes probable sensor drift for the microcats, as described in a previous section. The difference is more likely due to CTD calibration offset, as per the AM02 CTD salinity discussed above.



<u>Figure 1:</u> Schematic of AMISOR borehole microcat moorings. Vertical dimensions are at times of mooring deployments.

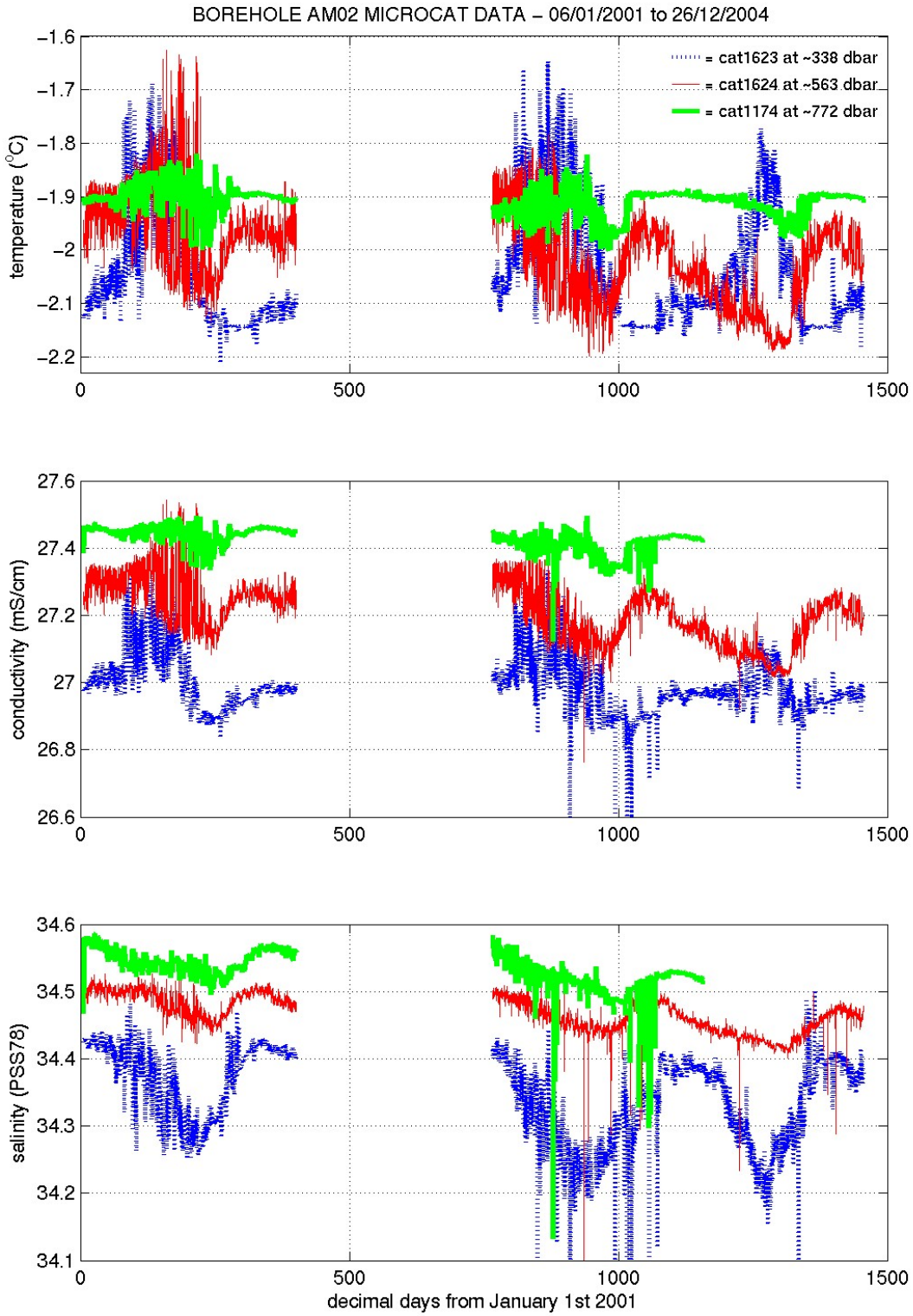


Figure 2: Borehole AM02 microcat data, January 2001 to December 2004.

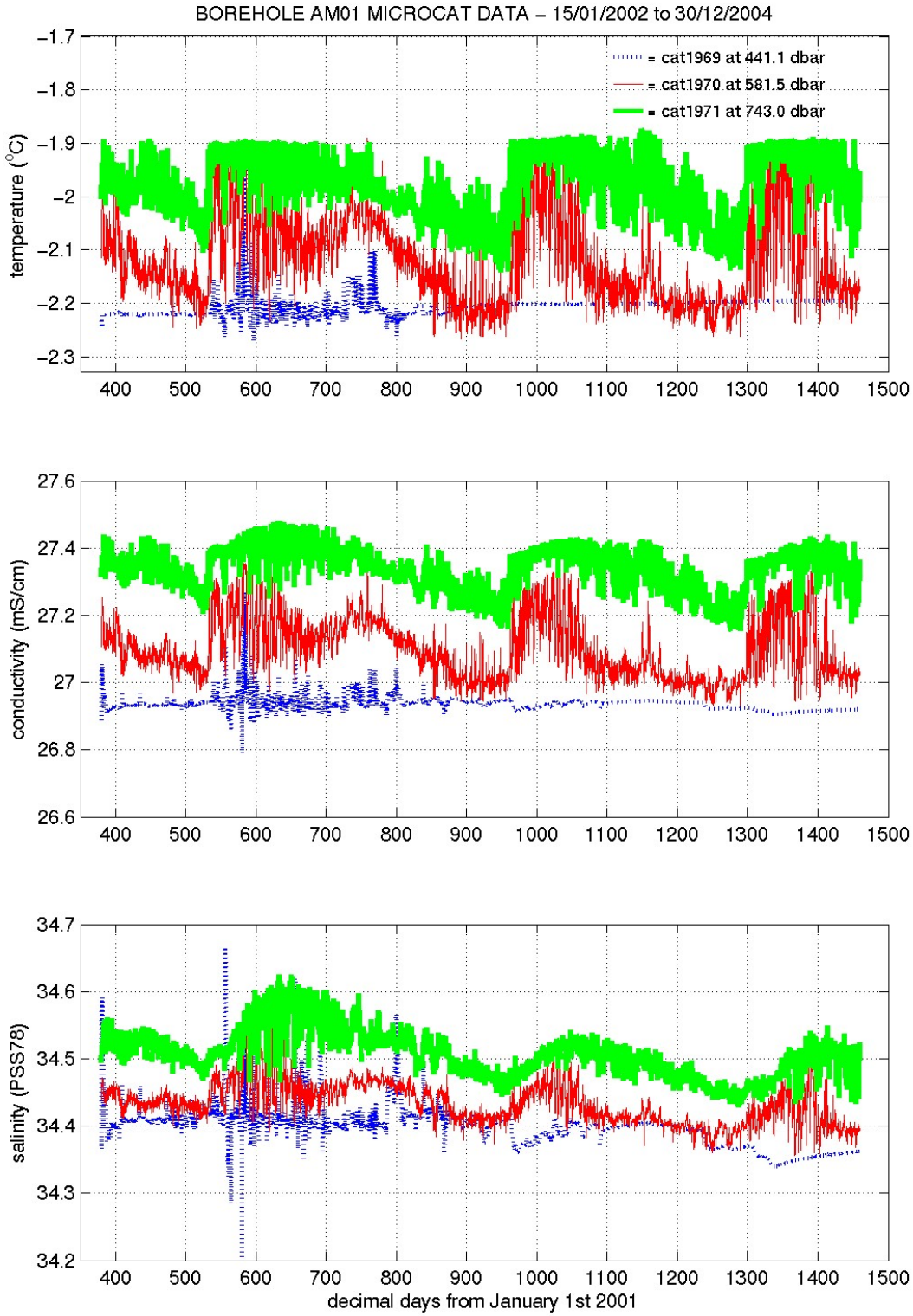


Figure 3: **Borehole AM01 microcat data, January 2002 to December 2004.**

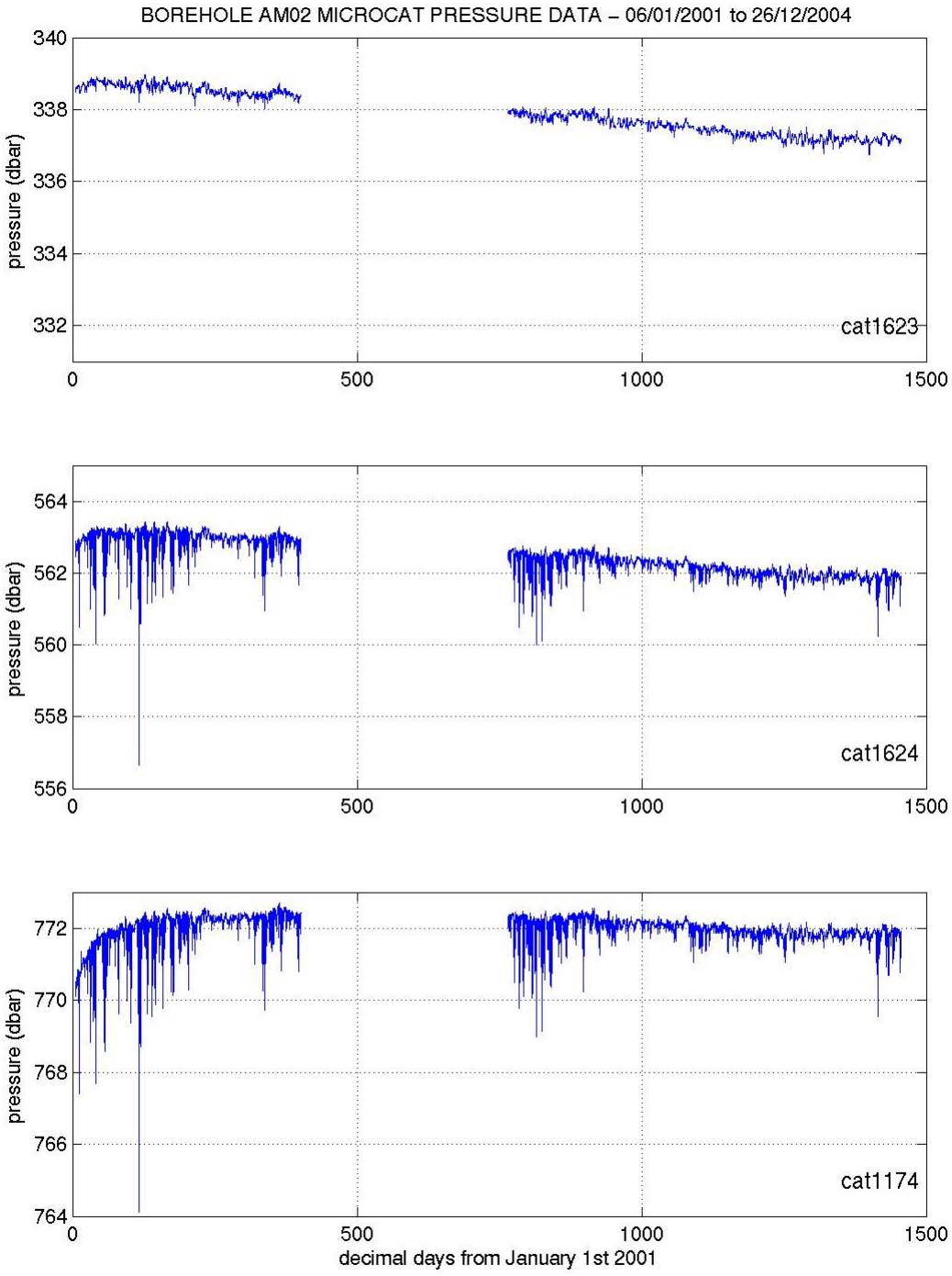


Figure 4: Borehole AM02 microcat pressure data. Note: vertical scales are all the same.

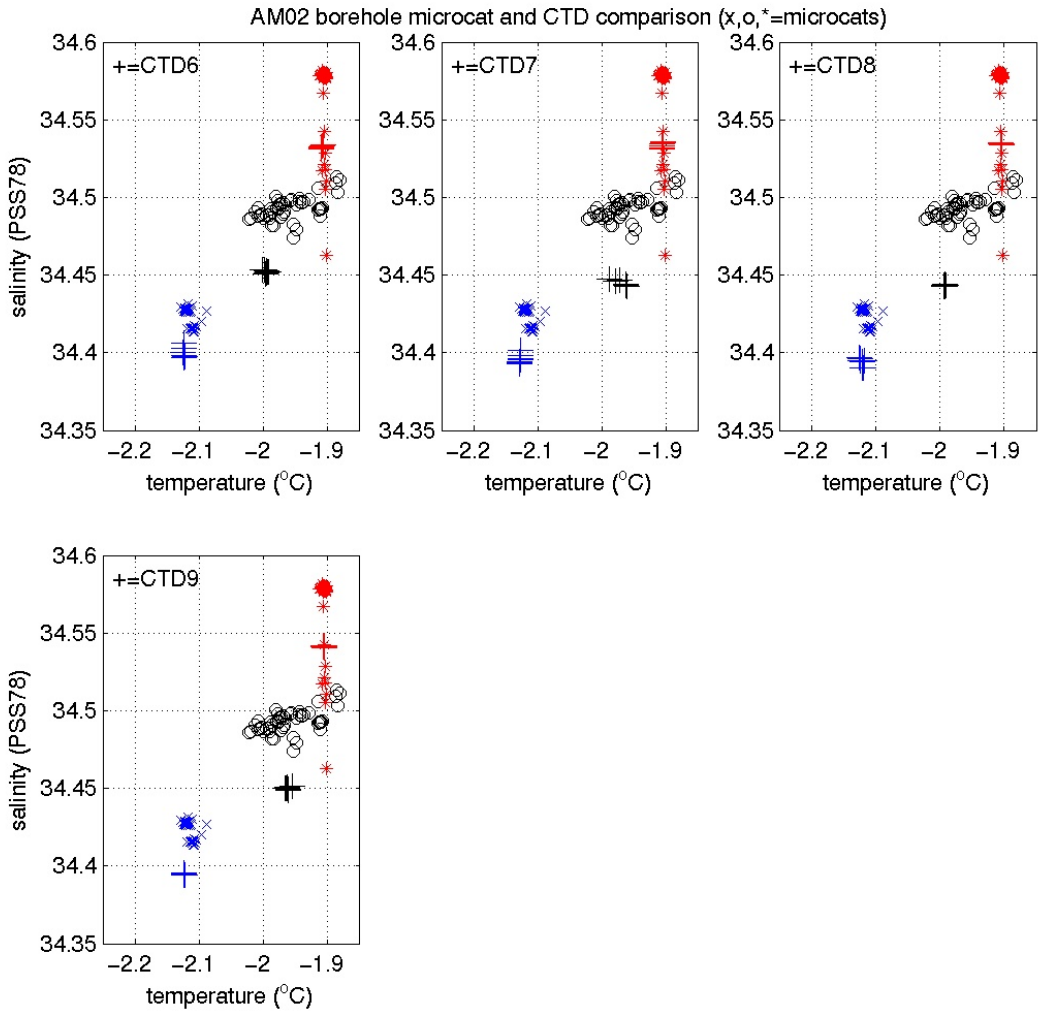
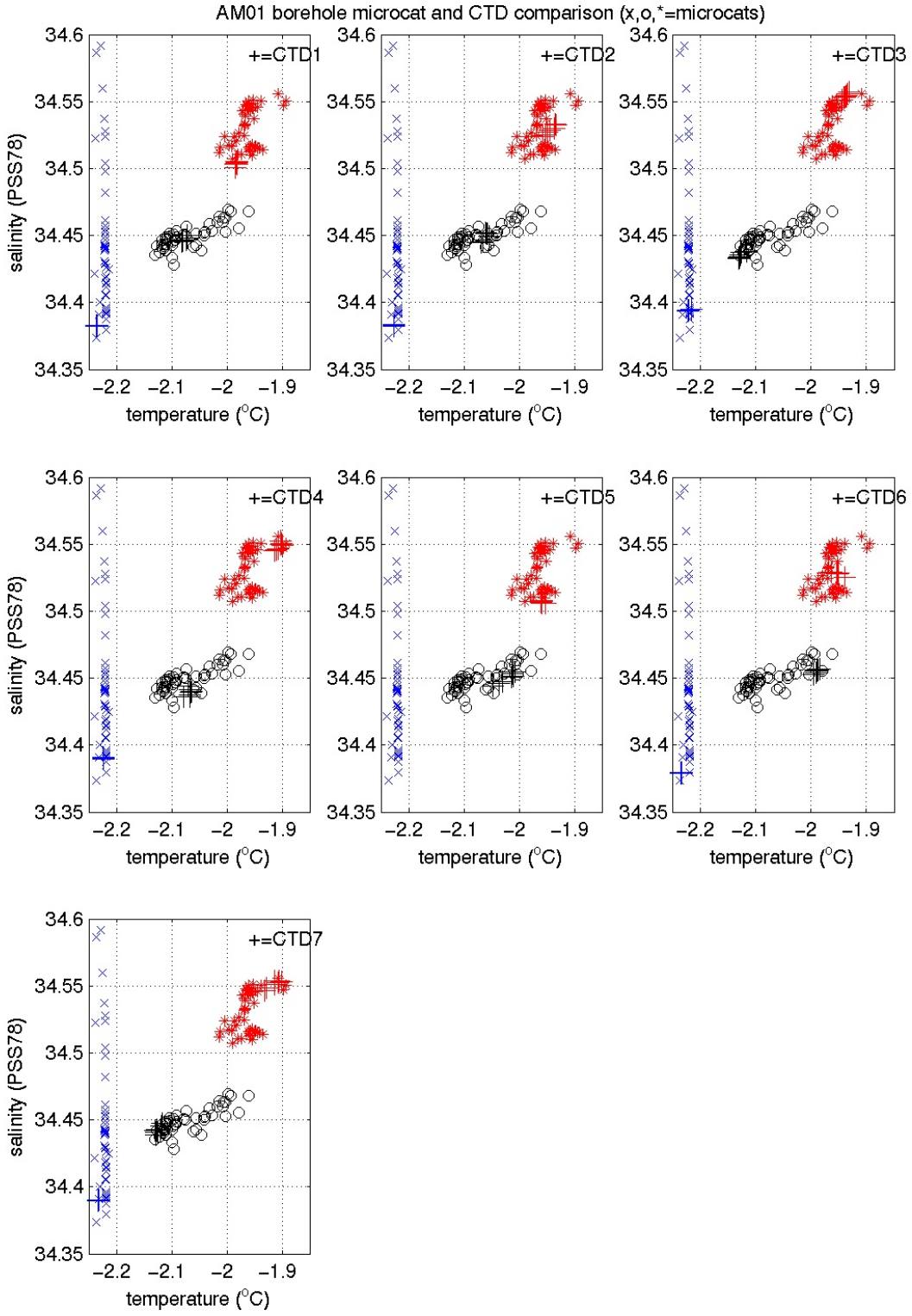
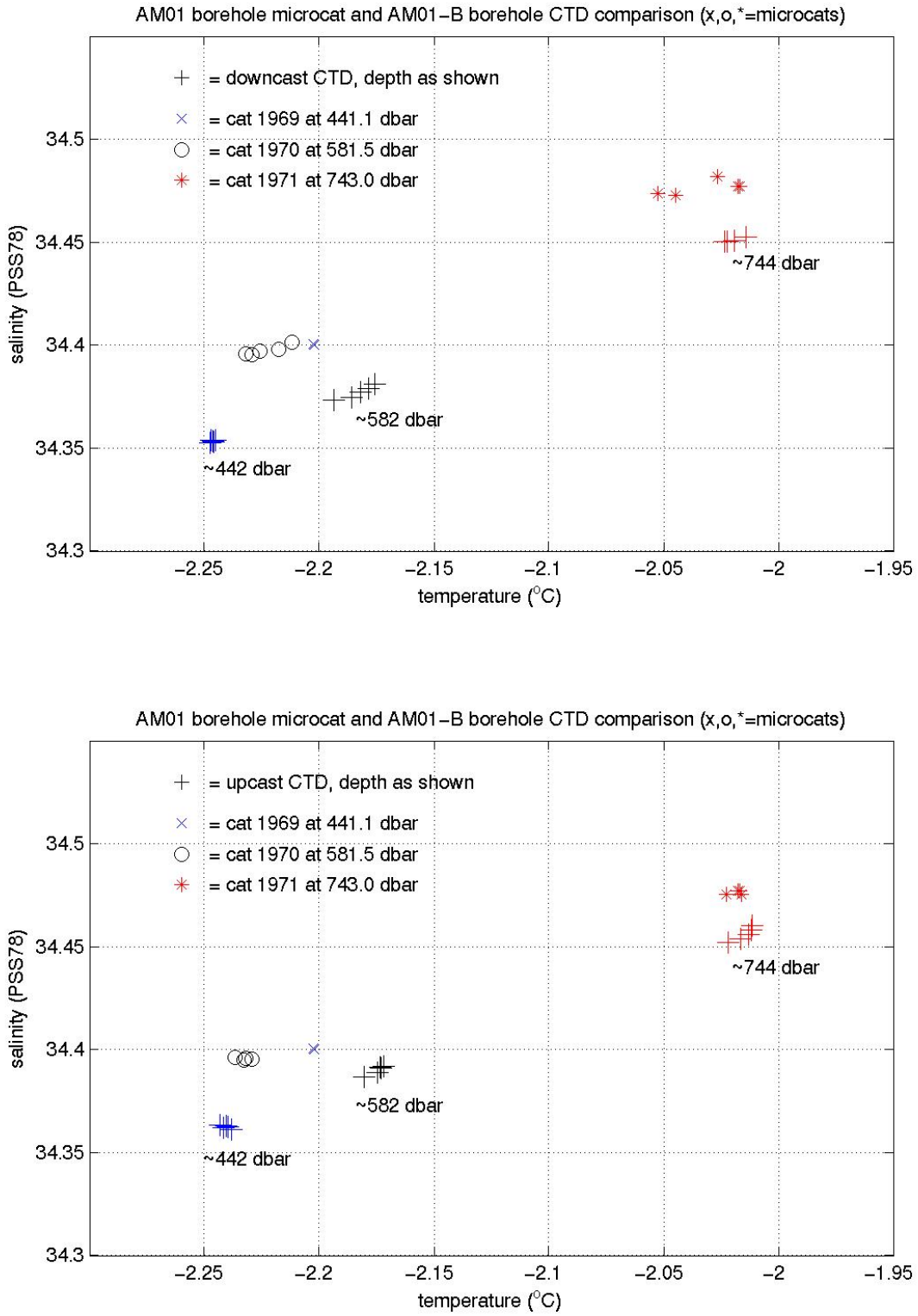


Figure 5: Borehole AM02 (2000/2001 season) – comparison of CTD and microcat data at the depth of the microcats. Note: downcast CTD data; 8 days of microcat data.



<u>Figure 6:</u> Borehole AM01 (2001/2002 season) – comparison of CTD and microcat data at the depth of the microcats. Note: downcast CTD data; 8 days of microcat data.



<u>Figure 7:</u> Comparison of CTD data from borehole AM01-B, and microcat data from borehole AM01, at the depth of the microcats. Season 2003/2004 data. Note: top panel shows downcast CTD data; bottom panel shows upcast CTD data; 2 hours of microcat data.

4. CTD DATA

4.1 General

A total of 16 CTD casts, using FSI 3" MicroCTD serial 1610, have been taken to date through the AMISOR boreholes: 8 through borehole AM02 during the 2000/2001 season; 7 through borehole AM01 during the 2001/2002 season; and a single cast through borehole AM01-B, approximately 1.6 km northeast of AM01, during the 2003/2004 season (Table 2). Following the ice shelf field work in 2000/2001, calibration checks of the borehole FSI MicroCTD were performed on 2 CTD casts aboard Aurora Australis cruise au0106 (Table 2), en route back to Hobart. Similarly, following the 2001/2002 field work, calibration checks were performed on 3 CTD casts on cruise au0207 (Table 2) en route back to Hobart (see Rosenberg et al., 2002, for complete information on marine science cruises).

Temperature, pressure and conductivity data are output from the borehole CTD in engineering units, with the following laboratory calibration coefficients applied:
May 2000 manufacturer supplied coefficients for 2000/2001 season data;
August 2001 CSIRO (Division of Marine Research) supplied coefficients for 2001/2002 season data;
September 2003 CSIRO (Division of Marine Research) supplied coefficients for 2003/2004 data.
With these calibrations alone the data are not sufficiently accurate to be useful, and further calibration steps are required, as detailed below.

Table 2: Station details for AMISOR borehole CTD's, and Aurora Australis calibration CTD's from cruises au0106 and au0207.
Notes:
* **depth to water surface=distance from top of borehole down to water surface in the borehole**
*** bottom depth=total water depth from water surface to ocean bottom**
*** max.P=maximum pressure of CTD cast**
*** elevation=CTD elevation above bottom at the bottom of the cast**
*** for AM01, borehole depth given is the depth to the base of the porous ice/slush layer below the solid ice shelf**

borehole	*cast no.*	*start time (UTC)*	*latitude*	*longitude*	*borehole depth (m)*	*depth to water surf. (m)*	*bottom depth (m)*	*max.P (dbar)*	*elev. (m)*
AM02	1	1010 01-JAN-2001	69°42.8'S	72°38.4'E	380	46	790	800	0
AM02	3	1925 01-JAN-2001	69°42.8'S	72°38.4'E	380	46	790	256	537
AM02	4	0632 02-JAN-2001	69°42.8'S	72°38.4'E	372	47	790	788	11
AM02	5	1622 02-JAN-2001	69°42.8'S	72°38.4'E	372	47	790	768	31
AM02	6	0728 03-JAN-2001	69°42.8'S	72°38.4'E	372	47	790	778	21
AM02	7	0122 04-JAN-2001	69°42.8'S	72°38.4'E	372	47	790	778	21
AM02	8	0817 05-JAN-2001	69°42.8'S	72°38.4'E	372	47.5	790	778	21
AM02	9	1253 05-JAN-2001	69°42.8'S	72°38.4'E	372	47.5	790	778	21
AM01	1	0928 10-JAN-2002	69°26.5'S	71°25.0'E	478	56.5	783	782	10
AM01	2	0110 11-JAN-2002	69°26.5'S	71°25.0'E	478	56.5	783	772	20
AM01	3	1748 11-JAN-2002	69°26.5'S	71°25.0'E	478	56.5	783	780	12
AM01	4	1136 12-JAN-2002	69°26.5'S	71°25.0'E	478	57.4	783	776	16
AM01	5	1531 12-JAN-2002	69°26.5'S	71°25.0'E	478	56.6	783	776	16
AM01	6	0902 13-JAN-2002	69°26.5'S	71°25.0'E	478	56.5	783	786	6
AM01	7	1143 14-JAN-2002	69°26.5'S	71°25.0'E	478	56	783	772	20
AM01-B	1	0348 24-Dec-2003	69°25.9'S	71°26.8'E	487	57	791	764	36
au0106 CTD									
	94	0243 28-FEB-2001	65°09.55'S	84°33.84'E			-	702	-
	95	0412 28-FEB-2001	65°09.68'S	84°34.04'E			-	2002	-
au0207 CTD									
	53	0152 27-FEB-2002	64°41.05'S	73°01.27'E			3490	2004	-
	54	0527 27-FEB-2002	64°33.13'S	73°36.04'E			3500	2004	-
	55	0739 27-FEB-2002	64°32.41'S	73°32.58'E			3500	1504	-

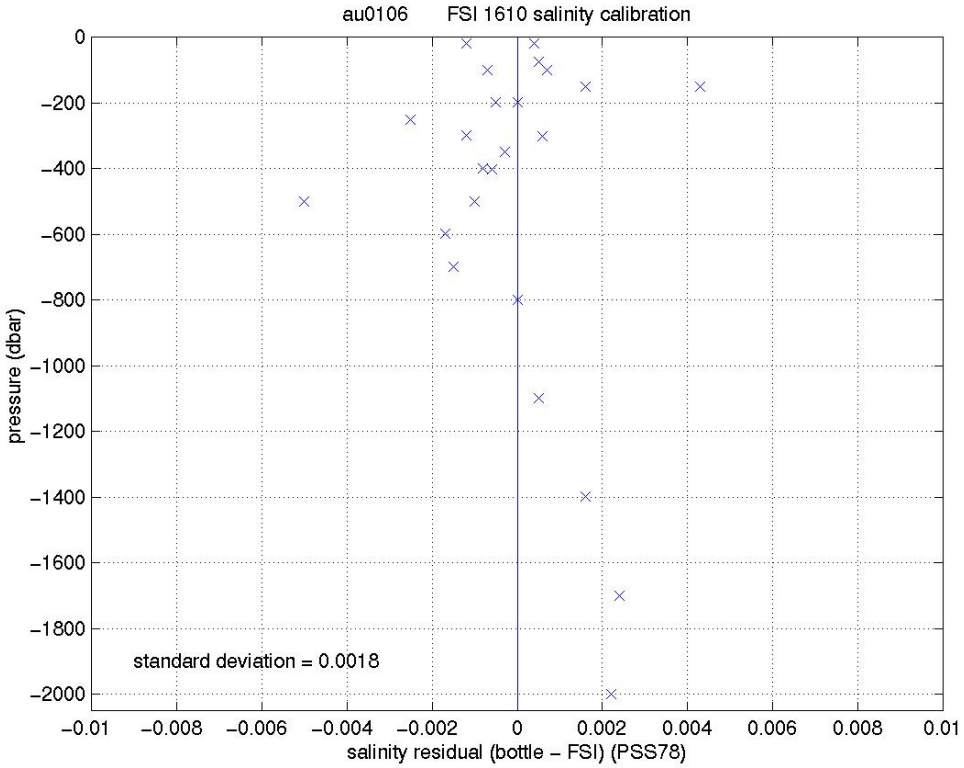


Figure 8: Salinity residual (bottle – FSI) for au0106 calibration check data, after application of ship-derived conductivity correction to FSI CTD data.

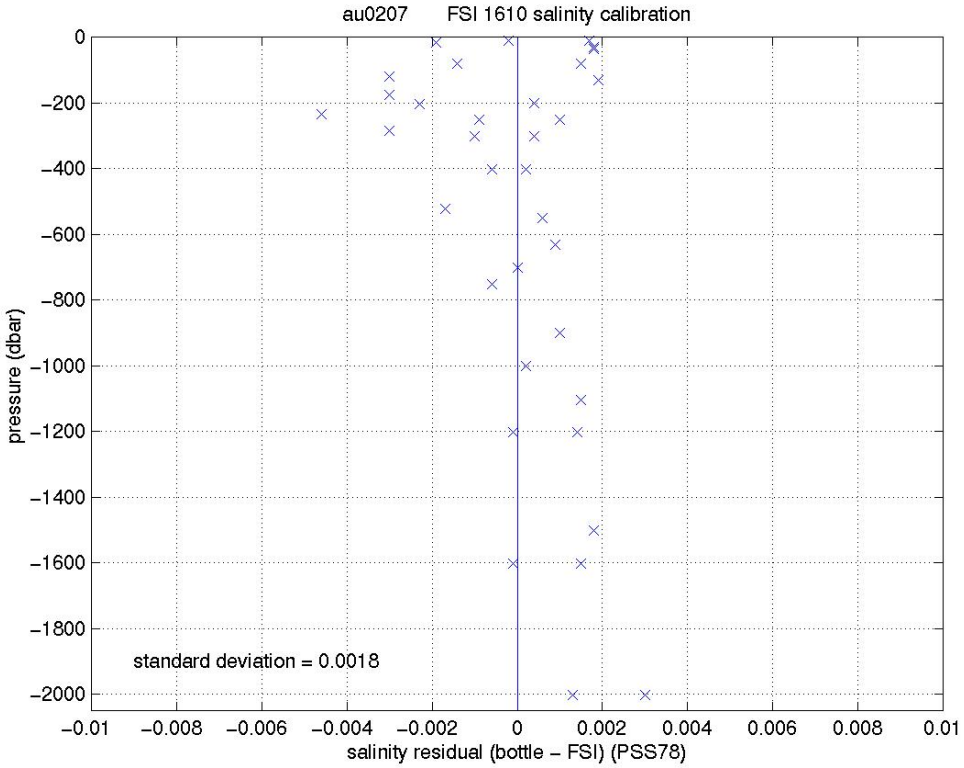


Figure 9: Salinity residual (bottle – FSI) for au0207 calibration check data, after application of ship-derived conductivity correction to FSI CTD data.

4.2 CTD data calibration and processing

2000/2001 season (AM02)
CTD conductivity calibrations are usually obtained using in situ salinity bottle samples. Unfortunately, the bottle samples collected through the borehole were not useful due to malfunction of the new Niskin bottle system deployed through the borehole along with the CTD. Final conductivity calibrations for the borehole data were therefore obtained from the 2 calibration casts aboard Aurora Australis cruise au0106. For these 2 casts the FSI MicroCTD, in internally recording battery-powered mode, was attached to the ship's main rosette system, and 2 routine 12 bottle casts were taken (Table 2) with GO (i.e. General Oceanics) CTD serial 1193.

The following processing steps were followed for the 2 au0106 casts to obtain calibration corrections for the FSI pressure and conductivity:

* Surface pressure offset was found by averaging the 20 pressure points previous to the CTD entering the water. This offset was then removed from FSI pressure data.

* Upcast burst data were formed by retaining the 30 sec. of data previous to each bottle firing, then averaging these 30 sec. bursts. Burst averages were then merged with GO upcast burst averages, and salinity bottle data.

* Separate pressure monotonic files were formed for downcast and upcast data.

* Comparison of FSI and GO pressure data revealed a small calibration difference, of the order 6 dbar over 2000 dbar. Assuming GO pressure as the more accurate, a correction was found for FSI pressure as follows. The upcast pressure burst averages for the FSI CTD were linearly fitted to the GO pressure burst averages. The following linear correction was then applied to all FSI pressure data:

$$p_{cal} = 1.00306\ p_{raw} + 0.24599 \qquad \text{(eqn 1)}$$

where p_{cal} and p_{raw} are respectively the corrected and uncorrected FSI pressure. Note that when obtaining the best fit, equal weight was given to both a fit through 0 pressure at the surface, and to the rest of the pressure data. However, application of this pressure correction still causes a small error of ~0.3 dbar to pressures near the surface.

* FSI conductivity was calibrated using the shipboard salinity bottle data (Figure 8), as per the method described in Rosenberg et al. (1995). Both stations were grouped together to provide a single calibration fit (i.e. no station dependent term). The linear correction obtained was:

$$c_{cal} = 0.99192\ c_{raw} + 0.080047 \qquad \text{(eqn 2)}$$

where c_{cal} and c_{raw} are respectively the corrected and uncorrected FSI conductivity; this correction was applied to all FSI conductivity data.

* 2 dbar averages were formed for temperature, corrected pressure and corrected conductivity, from the pressure monotonic downcast and upcast files. Note that a minimum attendance of 2 data points was required to form each 2 dbar bin. A salinity value for each 2 dbar bin was then calculated from these averages.

After the above processing for the au0106 FSI CTD data, CTD data from borehole AM02 were completed with the following processing steps:

* Data logged as station 2 was the upcast for station 1, and was appended to station 1 data (therefore no station 2) (Table 2).

* Surface pressure offsets were found and applied as described above.

* Separate pressure monotonic files were formed for downcast and upcast data, and the pressure and conductivity corrections found above (eqns 1 and 2) were applied.

* 2 dbar averaged files were formed for downcast and upcast data, as described above. Note that for the borehole data, a minimum attendance of 3 data points was required to form each 2 dbar bin.

* Note: touched bottom during station 1 - upcast data not affected.

2001/2002 season (AM01)

For conductivity, the initial correction for the borehole data was obtained from the 3 calibration casts aboard Aurora Australis cruise au0207. In this case, the ship's CTD was GO CTD serial 2568. FSI and GO CTD data were compared, FSI conductivity data were calibrated against the bottle samples obtained, and pressure corrections were applied, as described above for cruise au0106 data, with the following differences:

* The pressure calibration difference between the FSI and GO CTD's was of the order 4 dbar, and the linear correction applied to the FSI pressure data was:

$$p_{cal} = 1.0020\ p_{raw} + 0.2506 \quad \text{(eqn 3)}$$

* The linear calibration correction for the FSI conductivity, derived from the shipboard salinity bottle data (Figure 9), was:

$$c_{cal} = 0.99662\ c_{raw} + 0.080084 \quad \text{(eqn 4)}$$

Good salinity samples were obtained from the Niskin bottles deployed through borehole AM01 with the CTD, allowing further conductivity correction of the FSI borehole data. These borehole salinity samples were analysed on the ship on the return to Hobart. After the above processing for the au0207 FSI CTD data, further corrections and processing steps were carried out for CTD data from borehole AM01, as follows:

* Communication problems up the seacable were encountered when deploying the CTD through the borehole, and all stations were logged at ~0.3Hz. Note that the CTD was lowered and raised at slower rates than the previous season, to compensate for the decreased data frequency. For stations 6 and 7, the data were logged internally at 1.83 Hz. These internally logged data, at the higher sampling rate, were used for stations 6 and 7.

* Surface pressure offsets were found and applied, as described for the previous season.

* Separate pressure monotonic files were formed for downcast and upcast data, and the pressure and conductivity corrections found from the ship comparisons (eqns 3 and 4) were applied.

* The physical mounting of the FSI CTD on the borehole seacable and on the ship's rosette frame in both cases resulted in physical objects lying within the interference range of the conductivity cell. As a consequence, the ship-based conductivity correction was not expected to give the most accurate conductivity data for the borehole measurements. Good salinity samples were however obtained from the Niskin bottles deployed through borehole AM01, allowing an additional correction to be applied to FSI conductivity data. Salinity ranges below the ice shelf were small enough (~0.2 PSS78, Figure 20) that a simple offset correction was adequate. Comparing CTD and bottle salinities, the following offset correction was obtained:

$$c_{newcal} = c_{cal} + 0.0205 \quad \text{(eqn 5)}$$

where c_{cal} is the conductivity from equation 4 above, and c_{newcal} is the final corrected conductivity value (equivalent to a salinity correction of ~0.028 PSS78). This final correction was applied to all borehole AM01 CTD conductivity data.

* 2 dbar averaged files were formed for downcast and upcast data, as described for the previous season, and including calculation of salinity. Note that for the borehole data, a minimum attendance of 1 data point was required to form each 2 dbar bin.

* Complete CTD data (i.e not averaged into 2 dbar bins) for the time series station (logged as station 3a) are in the file fsiam0103a.cc

* Complete CTD data (i.e. not averaged into 2 dbar bins) for station 7, including 2 partial down and upcasts, plus stops at several depths for current measurements, are in the file fsiam0107a.cc

* Note: touched bottom during station 6 - upcast data not affected.

2003/2004 season (AM01-B)
A single CTD cast only was taken in this season, and with no associated ship-board calibration CTD's. Separate pressure monotonic files for downcast and upcast data were formed from the raw data. 2 dbar averages were then formed for temperature, pressure and conductivity from the pressure monotonic files. A minimum attendance of 3 data points was required to form each 2 dbar bin. A salinity value for each 2 dbar bin was then calculated from these averages.

Instrument communication problems were encountered during the CTD cast, and as a result some data were lost, including the first part of the downcast from the surface down to 363 dbar. Loss of the first part of the cast data meant that no pre-cast surface pressure offset was available. Thus no corrections were applied to the CTD pressure data, introducing a pressure error of ~0.5 dbar.

Salinity samples were collected from the 2 Niskin bottles, but were not analysed till 8 months after collection. The bottle-CTD salinity difference is inconsistent for the 2 samples (Table 3). In addition, the bottle salinities are significantly higher (by ~0.1 PSS78) than values from the AM01 site in season 2001/2002, and are higher than the nearby microcat values (Figure 7). The bottle salinities are therefore assumed to be suspect, so no further calibration correction could be applied to the CTD conductivity/salinity data.

Further CTD data processing notes are as follows:

* Data from digitiser channel 3 indicate the bottom contact switch was not working.

* Missing 2 dbar data: 2 - 362 dbar from downcast (data lost during instrument comm. problems)
724 - 726 dbar from upcast (data lost during instrument comm. problems)
2 - 332 dbar from upcast (bad data, most likely due to sensor freezing)

* Upcast conductivity and salinity data from 334 to 412 dbar are suspect. Unclear whether cause is (a) interference from ice crystals, or (b) data within a porous layer below the solid ice shelf - the lack of similar data on the downcast suggests (a) (Figure 23).

Table 3: Niskin bottle salinity samples versus CTD salinity values for AM01-B.

Niskin	pressure (dbar)	CTD salinity (PSS78)	bottle salinity (PSS78)
7	764	34.500	34.683
8	443	34.361	34.473

4.3 CTD data quality

4.3.1 Ship-based CTD comparisons

2000/2001 season: au0106 FSI and GO CTD comparisons
Data comparisons between the FSI and GO CTD's for the 2 calibration casts on cruise au0106 are shown in Figures 12 to 15.

From Figure 14, there is a temperature calibration difference between the two instruments, as follows:

above 0^oC $t_{fsi} > t_{GO}$ by ~0.003^oC

below -0.4^oC $t_{fsi} < t_{GO}$ by ~0.005^oC

-0.4 to 0^oC transition zone between above two ranges

There appears to be a calibration offset between the two instruments at positive temperatures; at sub-zero temperatures, the response of the two instruments is different. This comparison alone does not indicate which instrument is in error, and the FSI temperature data can therefore only be assumed accurate to 0.005^oC.

From Figure 15, FSI and GO CTD salinities compare well, to within ~0.002 (PSS78); the exception is the downcast data in the steep vertical gradients down to ~500 dbar, where FSI salinities are greater than GO values (Figure 15a).

2001/2002 season: au0207 FSI and GO CTD comparisons
Data comparisons between the FSI and GO CTD's for 1 of the 3 calibration casts on cruise au0207 are shown in Figures 16 to 18.

From Figure 17, the temperature calibration difference between the two instruments appears to be ~0.003^oC for the downcast, and ~0.005^oC for the upcast, with significantly greater differences at low temperatures around the temperature minimum (Figure 16). Closer inspection of the vertical temperature profiles for the two CTD's reveals the large temperature difference around the temperature minimum is in fact due to pressure calibration differences causing vertical offset of the two profiles. And the larger temperature difference apparent on the upcast (Figure 17b) is again due to pressure calibration differences – in this case there is hysteresis of the pressure sensor for one of the two CTD's, causing increased vertical offset of the upcast temperature profiles for the two instruments. So temperature values for the two CTD's agree to within 0.003^oC.

From Figure 18, FSI and GO CTD salinities compare reasonably well, to within ~0.003 (PSS78). As above for temperature, the pressure calibration differences exaggerate the salinity difference around steep vertical gradients.

4.3.2 Borehole CTD's

2000/2001 season: borehole AM02 FSI CTD data
Downcast FSI CTD data for borehole AM02 are shown in Figure 19. Note that data inside the borehole (i.e. top 300 dbar) are not shown in the figures.

Downcast and upcast temperature data agree well (not shown in figure), and all 8 stations are consistent for temperature (Figure 19). Note that at station 1, the CTD was accidentally laid on the bottom, however this does not appear to have affected temperature data.

Salinity data for stations 1 to 5 (Figure 19) are unrealistically high when compared to ship-based measurements from the region (Figure 21), thus station 1 to 5 salinity data are assumed to be bad. Note that conductivity values dropped after touching the bottom during station 1, however values soon returned to normal on the upcast. The precision of salinity data for stations 6 to 9 is good, with good agreement between downcast and upcast data (not shown in figure). Without salinity bottle data to provide in situ calibrations, these data cannot be considered up to the usual accuracy. In fact data from later seasons indicate that these salinities are low by ~0.03 (PSS78) (i.e. conductivity low by ~0.02 mS/cm). The reason for the anomalously high salinity (i.e. conductivity) data for stations 1 to 5 is not known, however likely causes are either a hardware fault, or else physical interference with the field surrounding the inductive conductivity cell (i.e. an object too close to the sensor, but no longer there after station 5).

2001/2002 season: borehole AM01 FSI CTD data

Downcast FSI CTD data for borehole AM01 are shown in Figure 20. Note that data inside the borehole (i.e. top 300 dbar) are not shown in the figures. Downcast and upcast temperature and salinity data agree well (not shown in the figure), and in general the data are good for all 7 stations. Application of the additional conductivity offset correction derived from comparison with the Niskin bottle salinity samples, as described above, makes the FSI salinity data more accurate than data from the previous season (2000/2001, borehole AM02).

The profiles (Figure 20) clearly show the transition between the solid ice shelf and the porous layer at ~325 dbar. The next transition between the porous layer and clear water can be seen at ~420 dbar. Most stations then show a fairly homogeneous layer of ice shelf water below this, with a layer thickness of between 50 and 80 dbar.

2003/2004 season: borehole AM01-B FSI CTD data

Downcast and upcast data are shown in Figure 23. As no reliable in situ bottle samples were collected, and no shipboard calibration checks were done, CTD salinity accuracy can only be estimated. From the experience of previous seasons, an accuracy value of ~0.03 PSS78 is likely. This is consistent with the microcat-CTD salinity differences in Figure 7.

Ship-based CTD data

Ship-based CTD data from cruises au0106 and au0207 along the Amery Ice Shelf front are shown in Figures 21 and 22. For au0106, only the stations closest to the borehole site (Figure 10) are plotted. For au0207, only stations 46 to 52 (Figure 11) are plotted.

Overall, these ship-based data provide qualitative confirmation of the borehole CTD data.

Summary of borehole CTD data quality

<u>AM02, 2000/2001:</u>
* data logged at 1.83 Hz
* data accuracy: temperature 0.005^{o}C (data good for stations 1 - 9)
  salinity low by ~0.03 (PSS78) (data good for stations 6 - 9, bad for stations 1 - 5)
  pressure ~2 dbar

<u>AM01, 2001/2002:</u>
* data logged at ~0.3 Hz for stations 1 to 5
* internally logged data, at the higher sampling rate of 1.83 Hz, used for stations 6 and 7
* good data for all 7 stations
* data accuracy: temperature < 0.005^{o}C
  salinity < 0.004 (PSS78)
  pressure ~2 dbar

<u>AM01-B, 2003/2004:</u>
* data logged at ~0.9 Hz
* good downcast data for 364 dbar to bottom
* good upcast data for 334 dbar to bottom (with missing data from 724 to 726 dbar)
* upcast conductivity/salinity data are suspect for 334 to 412 dbar
* assumed data accuracy: temperature < 0.005^{o}C
  salinity low by ~0.03 (PSS78)
  pressure ~2dbar

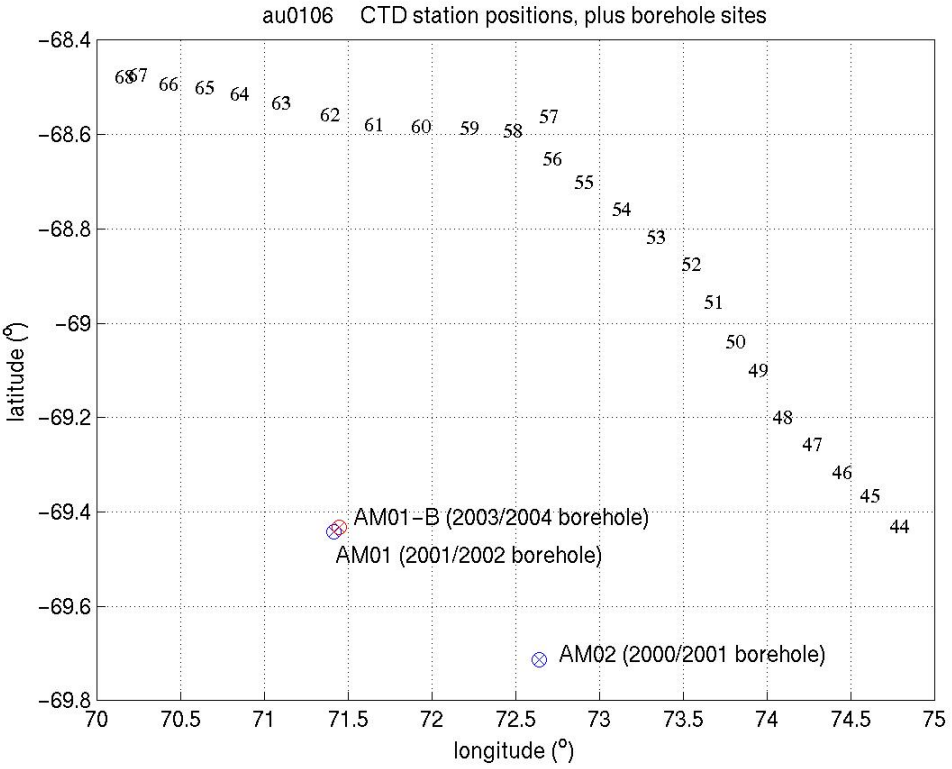


Figure 10: CTD station positions for cruise au0106 (ship-based AMISOR leg 1), and Amery Ice Shelf boreholes.

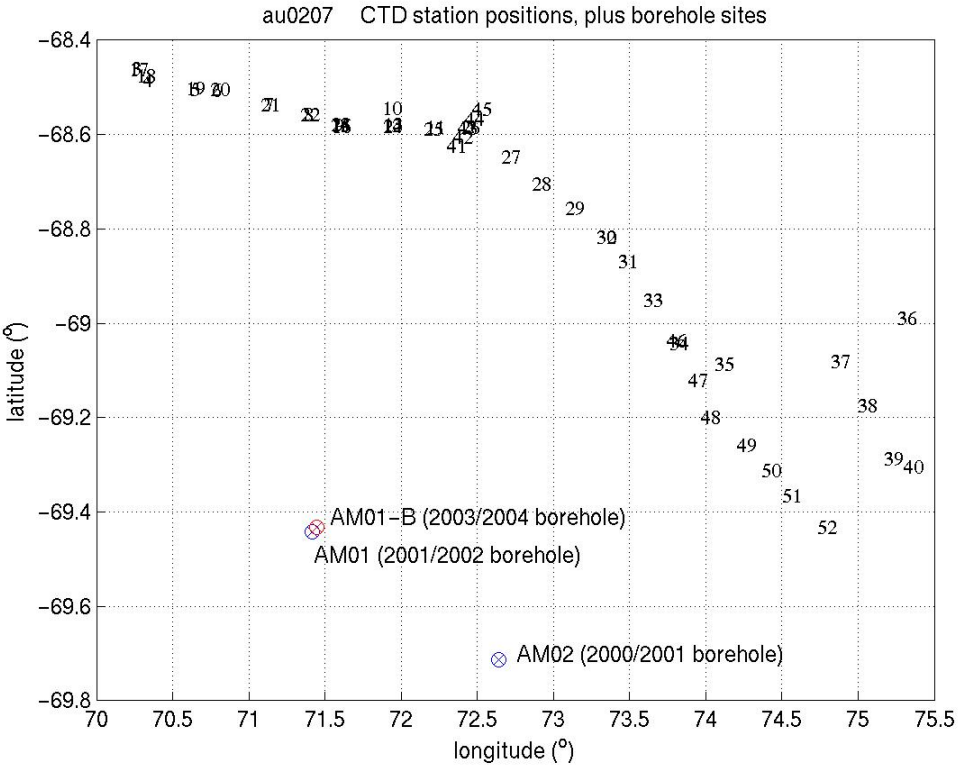


Figure 11: CTD station positions for cruise au0207 (ship-based AMISOR leg 2), and Amery Ice Shelf boreholes.

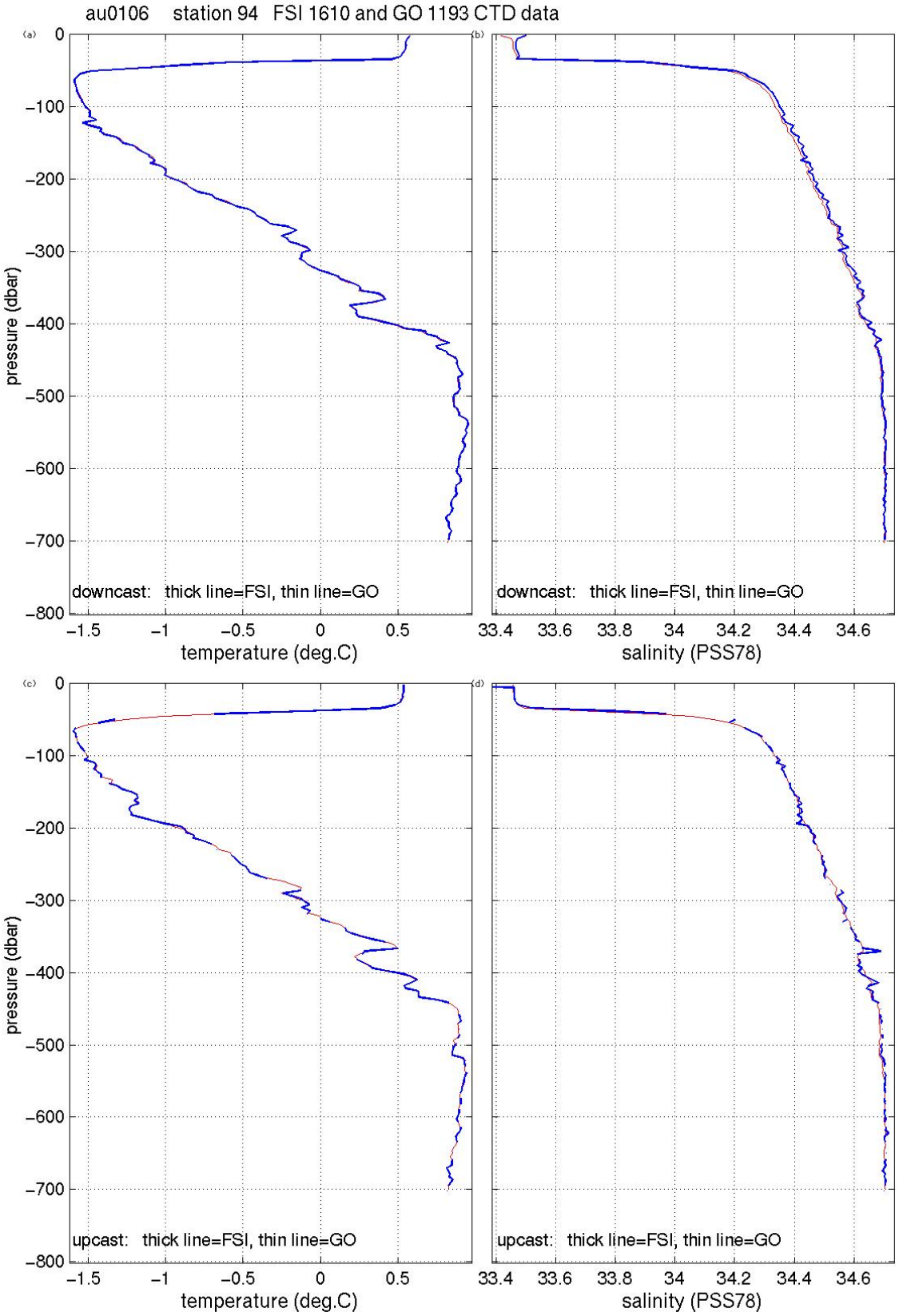


Figure 12: Comparison of FSI and GO CTD data, cruise au0106, station 94.

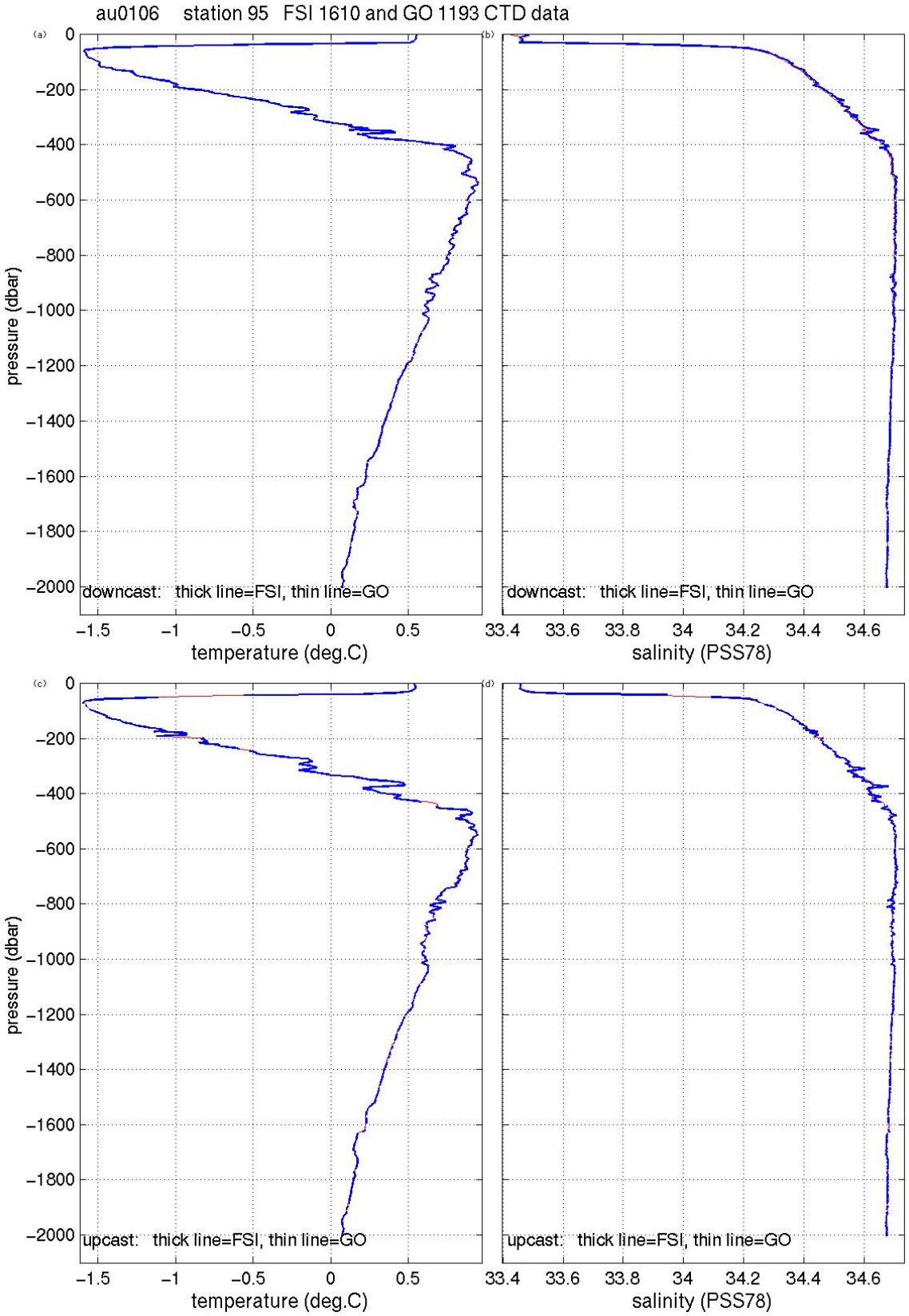


Figure 13: Comparison of FSI and GO CTD data, cruise au0106, station 95.

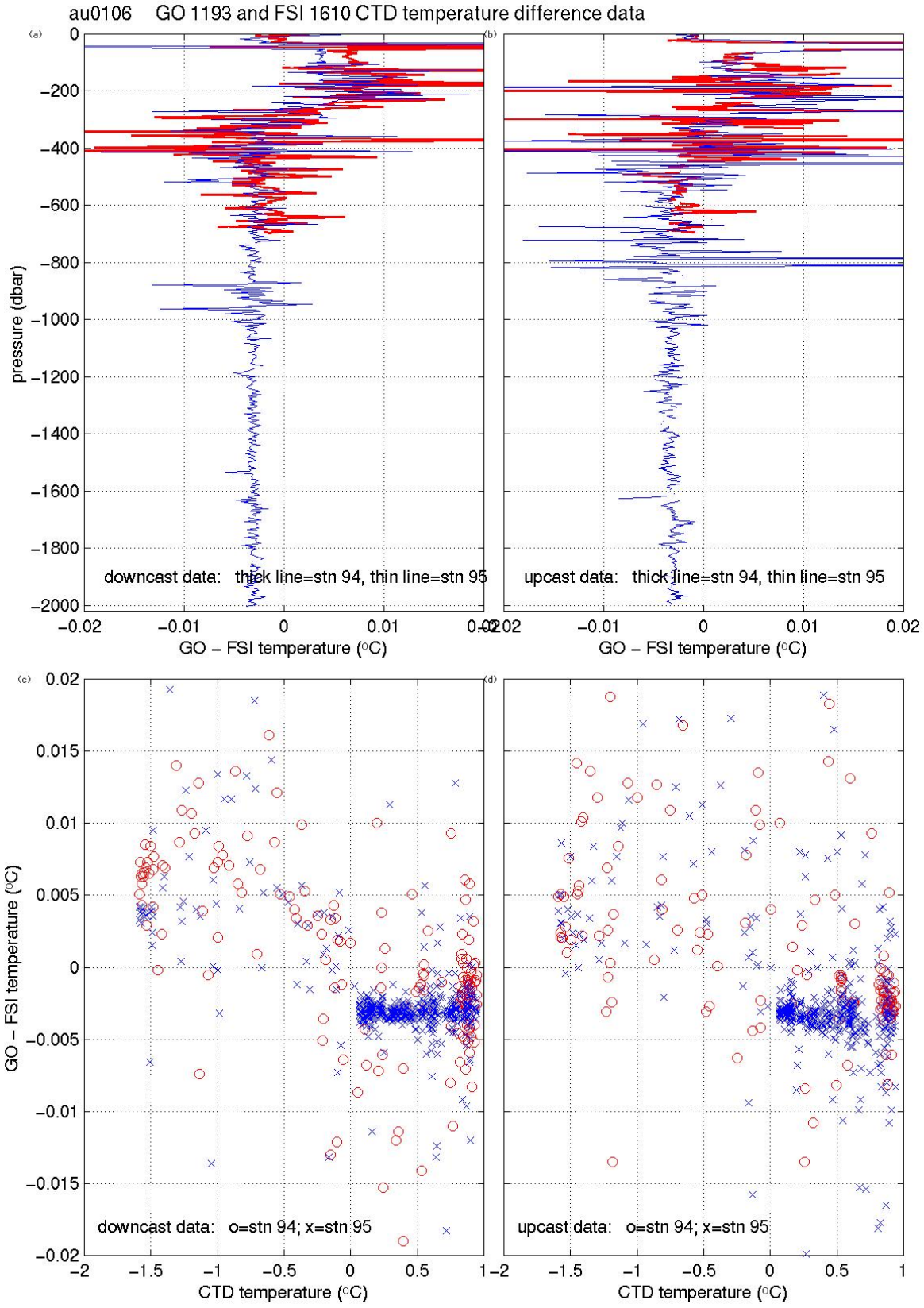


Figure 14: Difference between GO and FSI CTD temperature data, cruise au0106, stations 94 and 95. (a) difference for downcast data versus pressure; (b) difference for upcast data versus pressure; (c) difference for downcast data versus GO CTD temperature; (d) difference for upcast data versus GO CTD temperature.

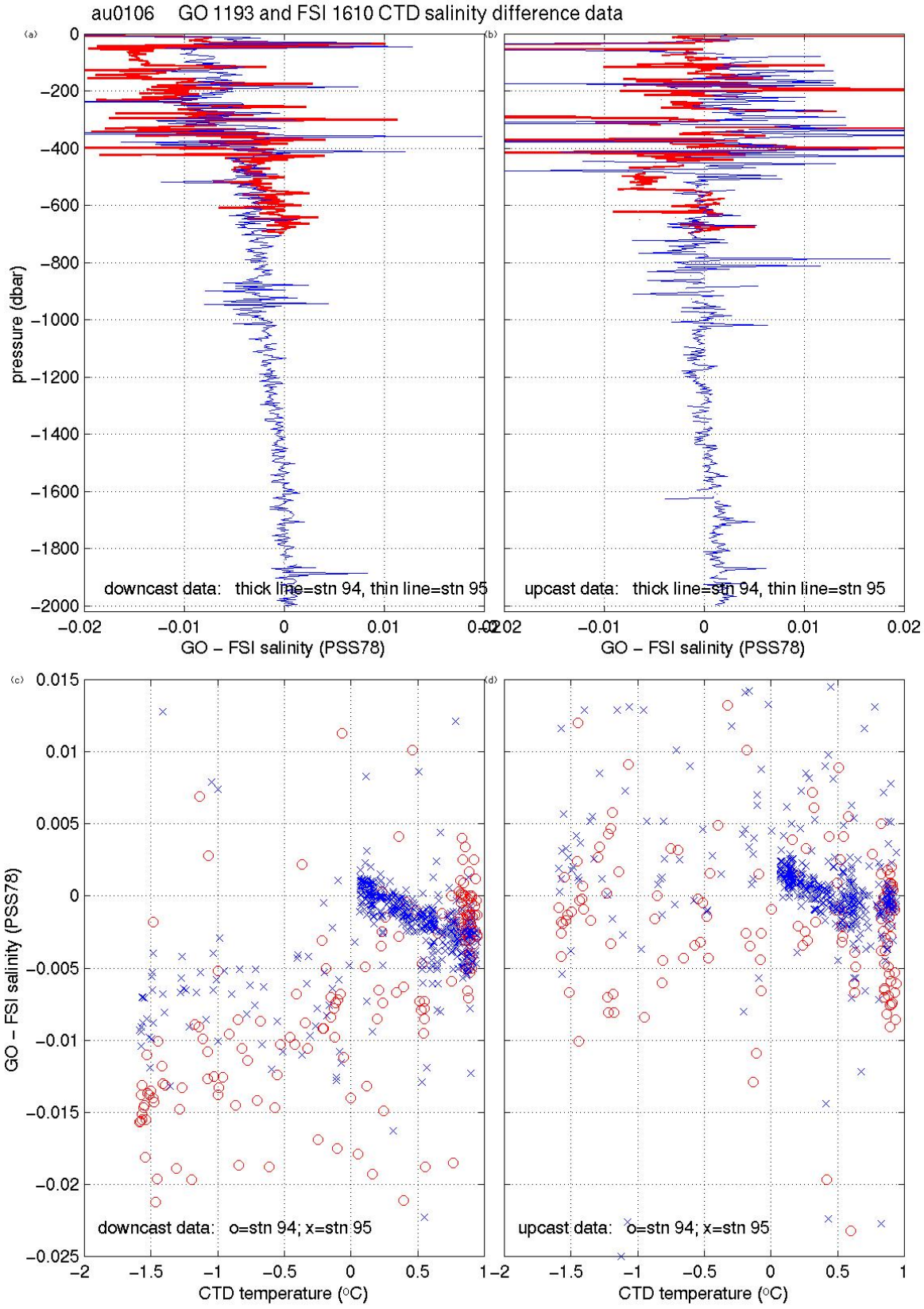


Figure 15: Difference between GO and FSI CTD salinity data, cruise au0106, stations 94 and 95. (a) difference for downcast data versus pressure; (b) difference for upcast data versus pressure; (c) difference for downcast data versus GO CTD temperature; (d) difference for upcast data versus GO CTD temperature.

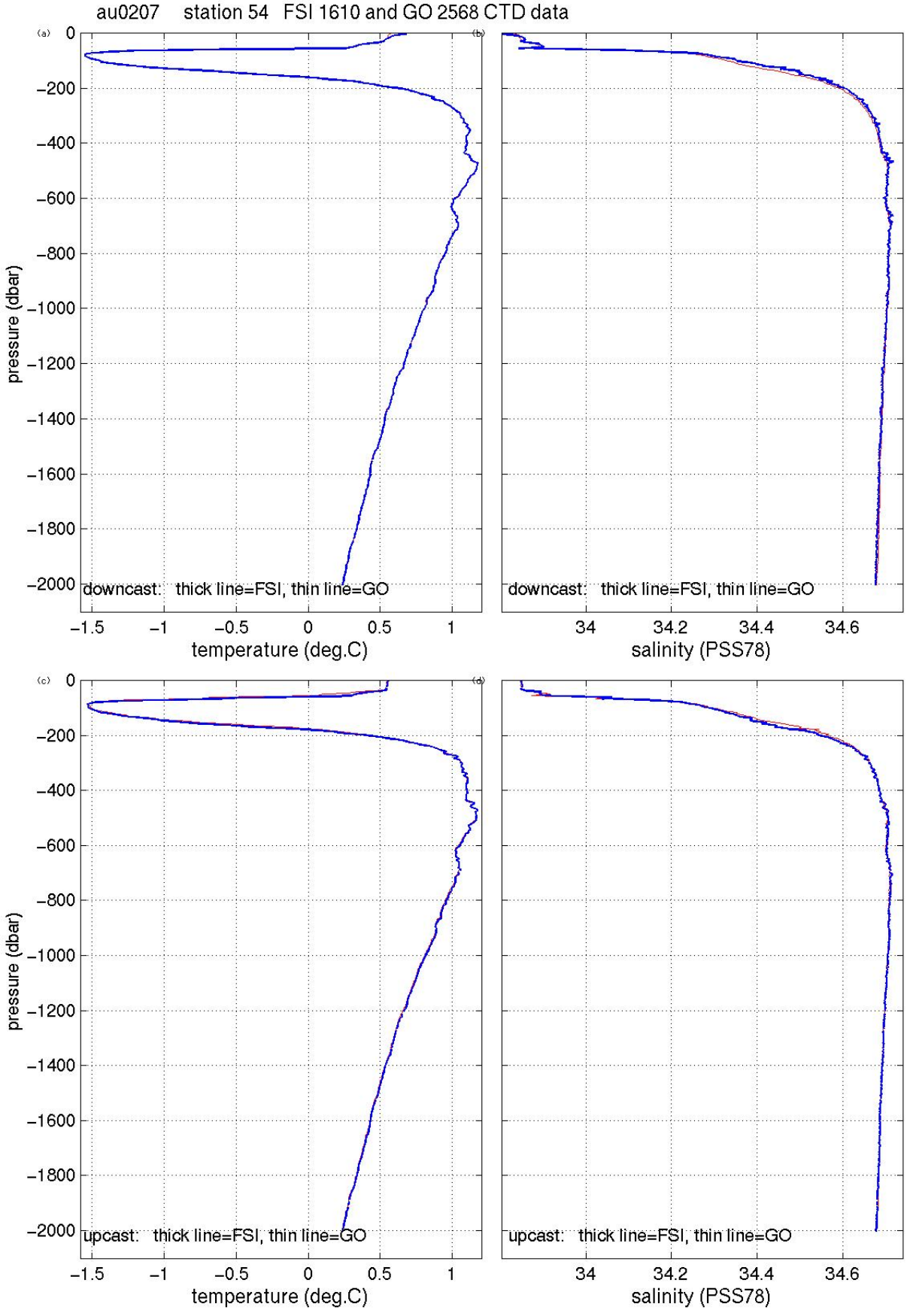


Figure 16: Comparison of FSI and GO CTD data, cruise au0207, station 54.

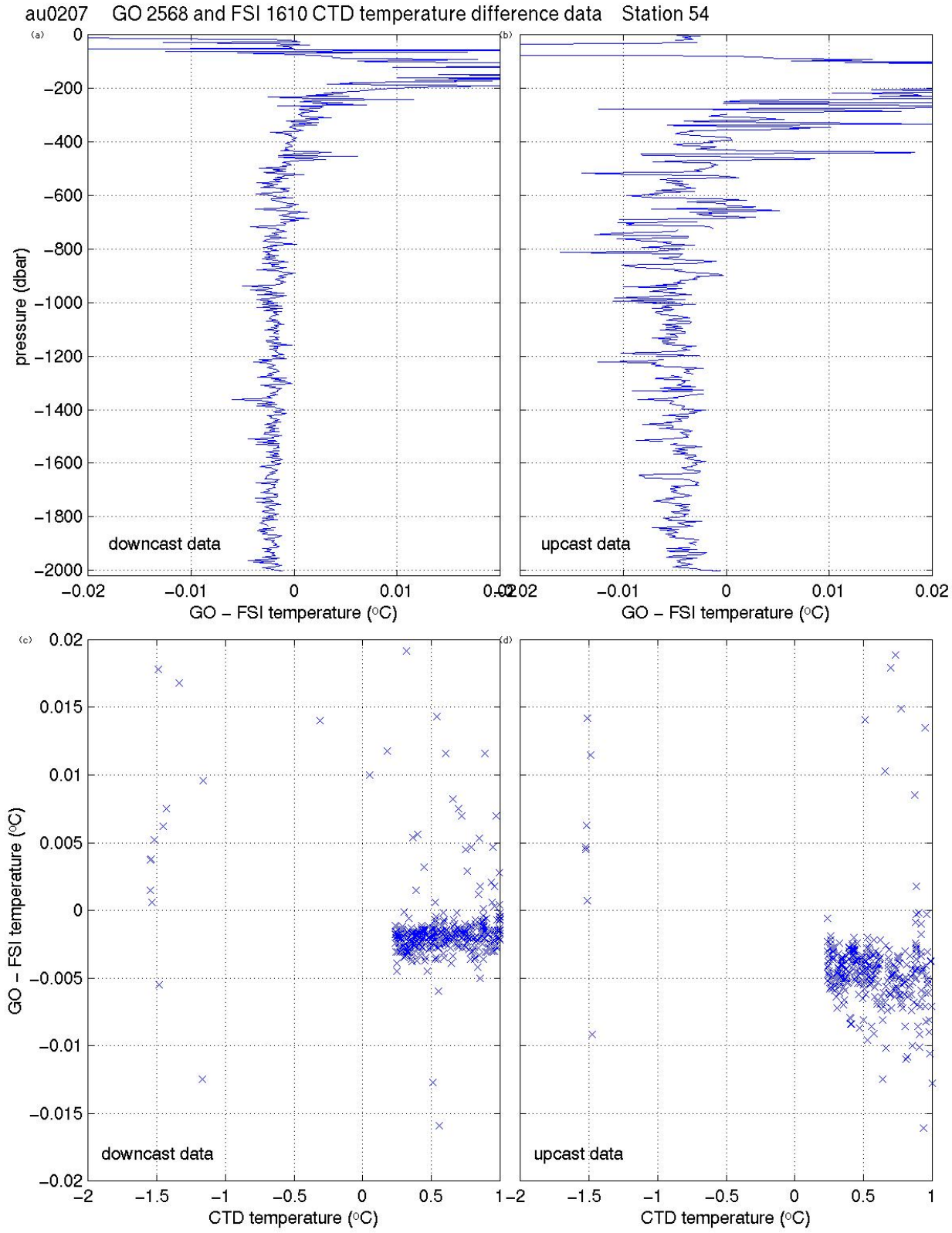


Figure 17: Difference between GO and FSI CTD temperature data, cruise au0207, station 54. (a) difference for downcast data versus pressure; (b) difference for upcast data versus pressure; (c) difference for downcast data versus GO CTD temperature; (d) difference for upcast data versus GO CTD temperature.

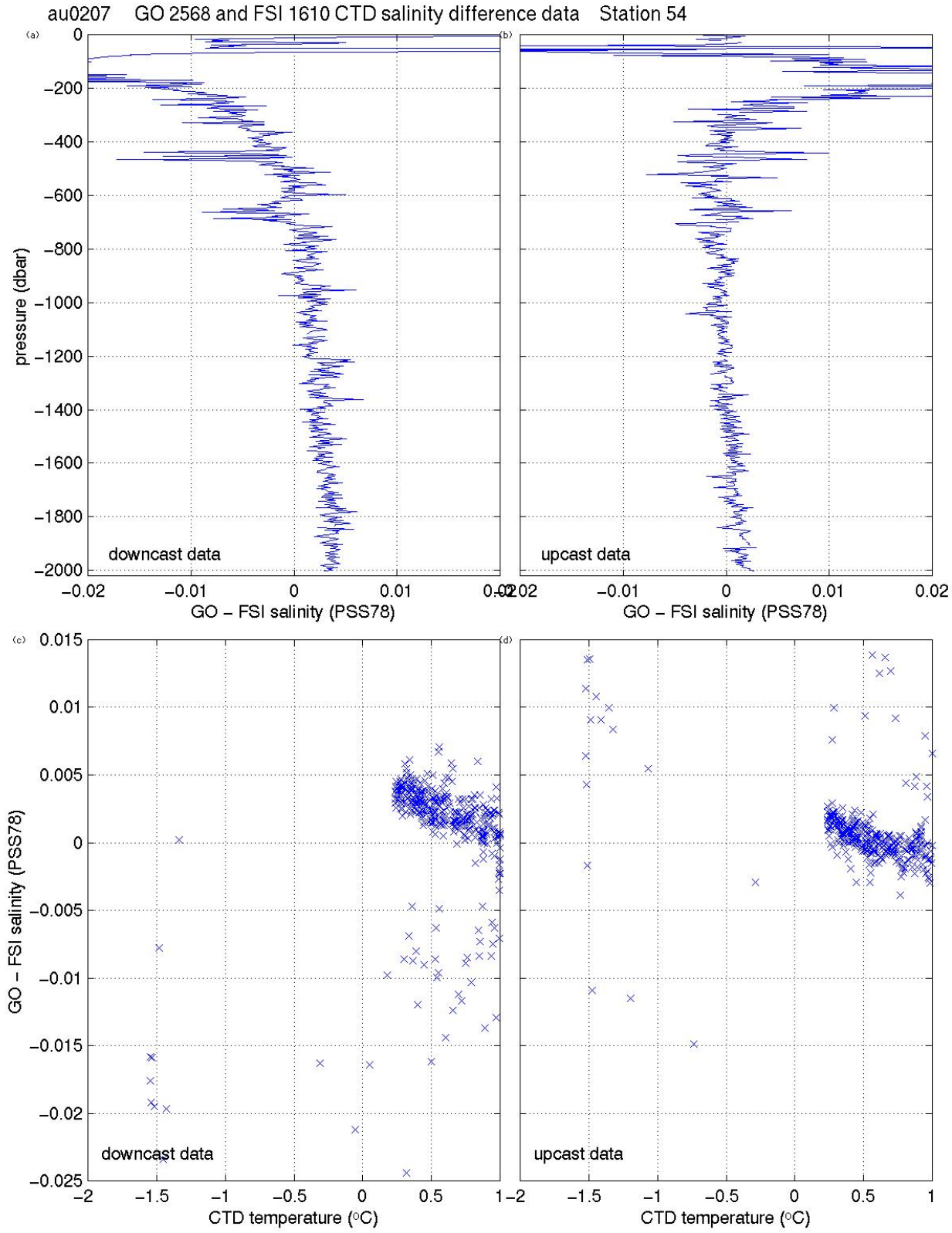
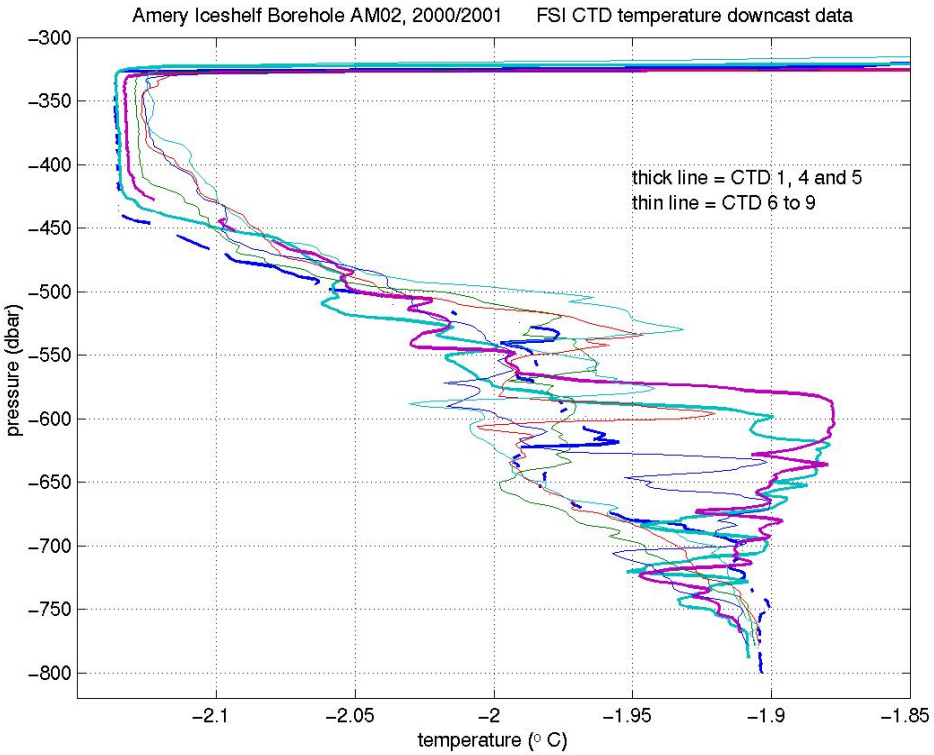
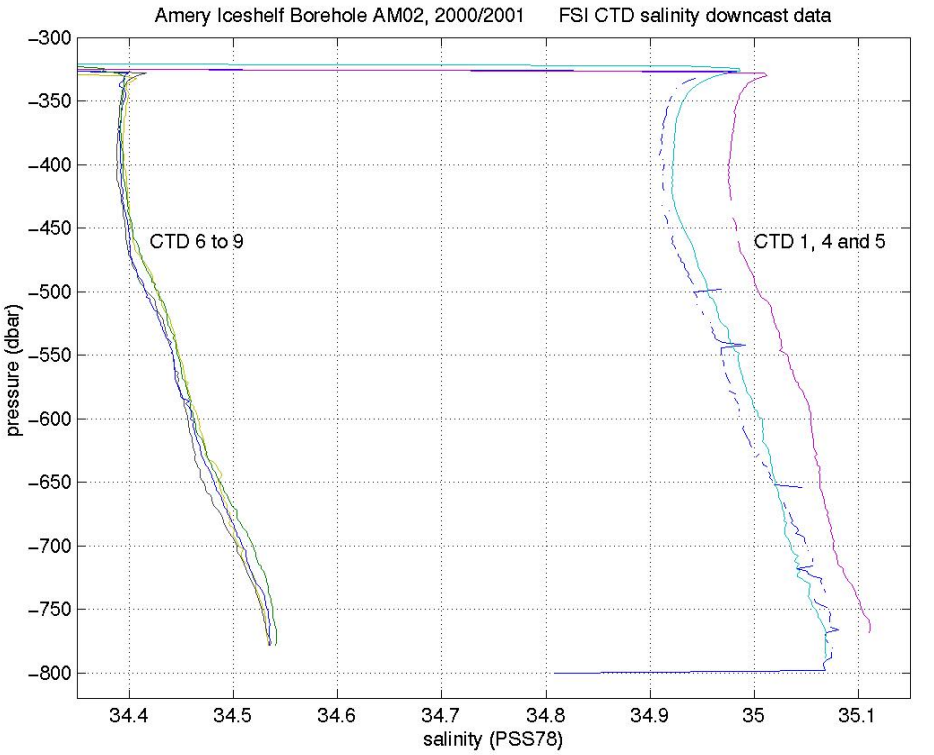
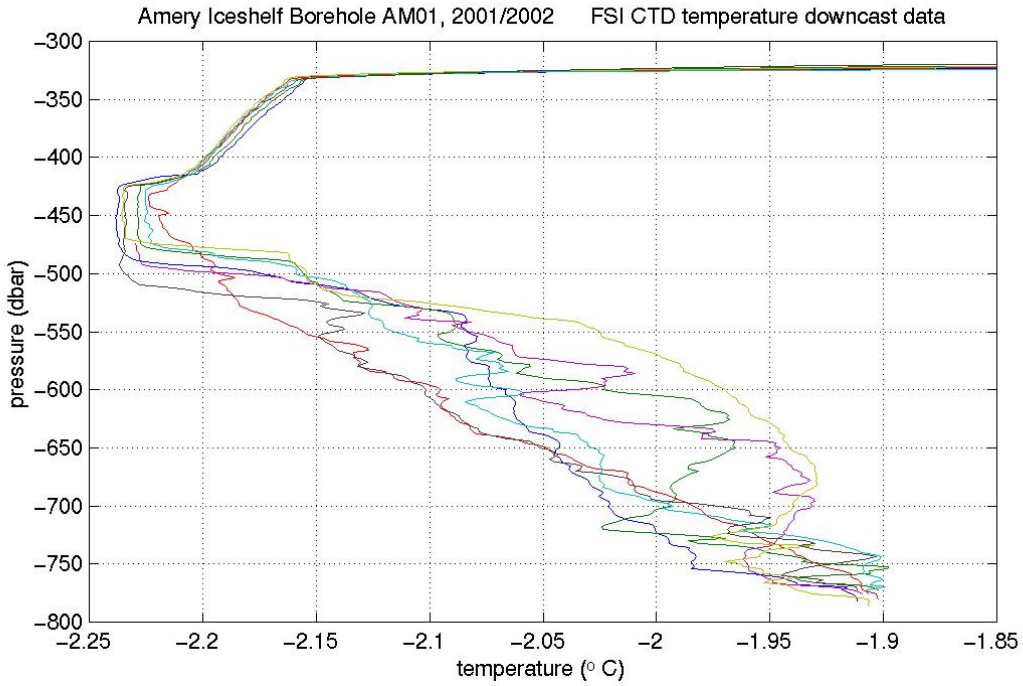


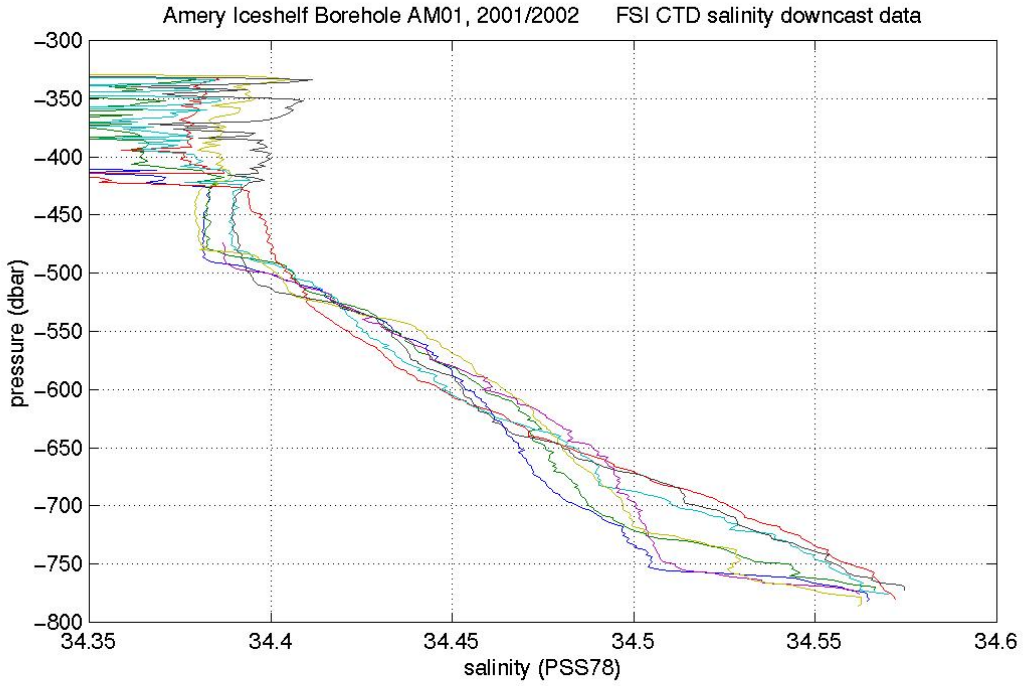
Figure 18: Difference between GO and FSI CTD salinity data, cruise au0207, station 54. (a) difference for downcast data versus pressure; (b) difference for upcast data versus pressure; (c) difference for downcast data versus GO CTD temperature; (d) difference for upcast data versus GO CTD temperature.





<u>Figure 19:</u> Amery Ice Shelf borehole AM02 CTD data: downcast temperature and salinity data below 300 dbar.





<u>Figure 20:</u> Amery Ice Shelf borehole AM01 CTD data: downcast temperature and salinity data below 300 dbar.

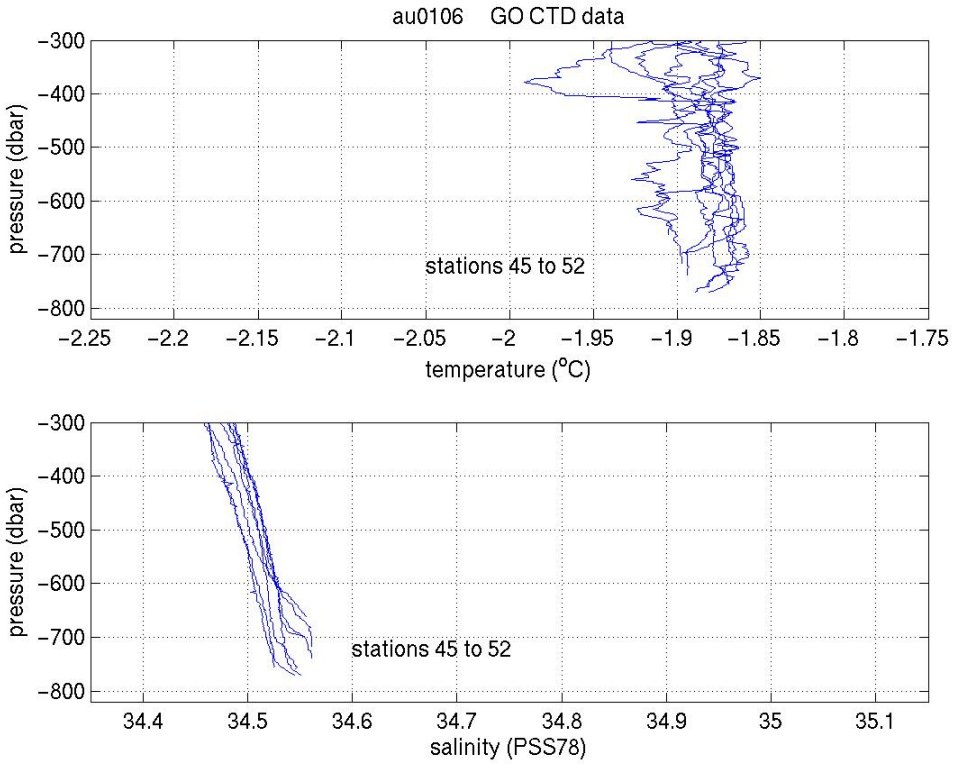


Figure 21: Cruise au0106 (ship-based AMISOR leg 1) CTD data: downcast temperature and salinity data below 300 dbar.

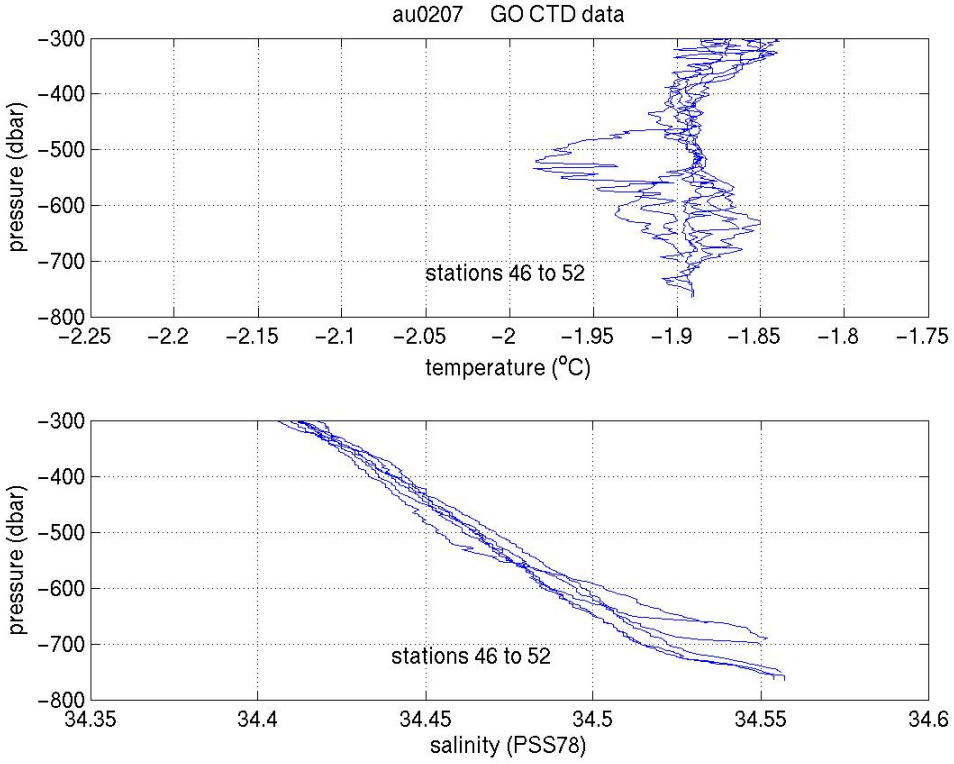
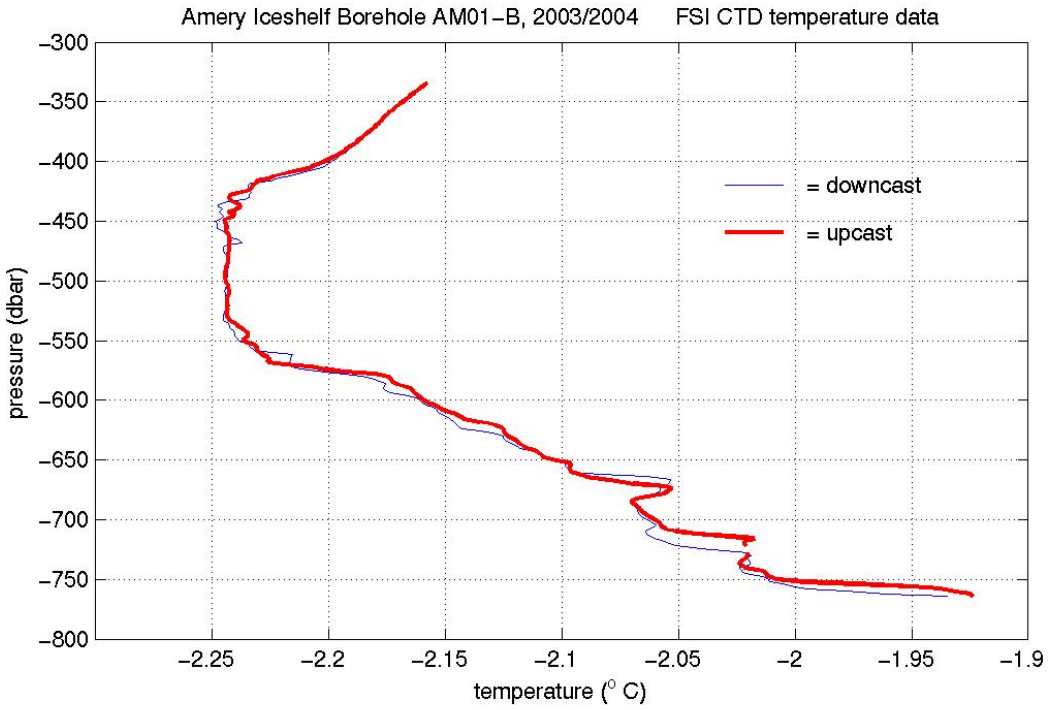


Figure 22: Cruise au0207 (ship-based AMISOR leg 2) CTD data: downcast temperature and salinity data below 300 dbar.



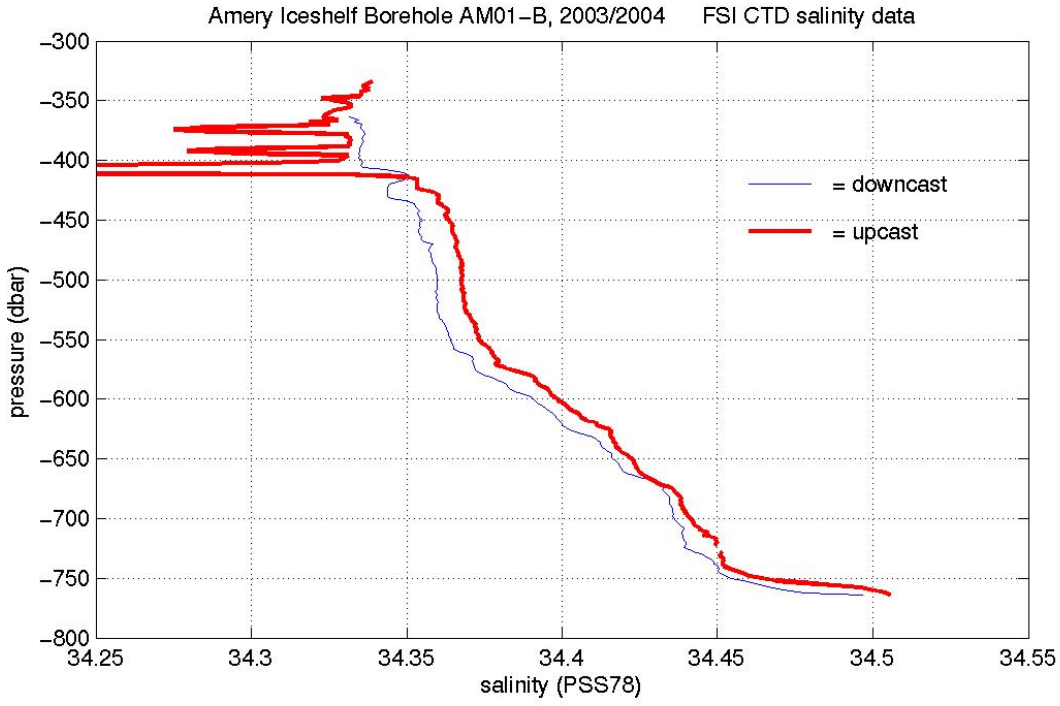


Figure 23: Amery Ice Shelf borehole AM01-B CTD 2 dbar averaged data.

5. DATA FILE FORMATS

5.1 Microcat data

Microcat data are provided in files for each download, and in files with all downloads joined into a single time series. Single time series data span the time intervals:
January 2001 to December 2004 for borehole AM02 (note: no data for 07/02/2002 to 04/02/2003)
January 2002 to December 2004 for borehole AM01

ascii
Microcat data from each download are contained in text files named catxxxx_y.dat, where:
xxxx = microcat serial number
y = download number
Each file contains 4 header lines, followed by the data in column format. The columns are:
1. decimal time (UTC) - decimal days from midnight on 31st December 2000. So, for example, midday on 15th January 2001 = decimal time 14.5 (corrected for clock drift, if applicable)
2. temperature (^{o}C, ITS90)
3. conductivity (mS/cm)
4. pressure (dbar)
5. salinity (PSS78)
6. to 11. day, month, year, hour, minute, second (UTC) (corrected for clock drift, if applicable)
12. decimal time (UTC) - no correction for clock drift
13. to 18. day, month, year, hour, minute, second (UTC) - no correction for clock drift

* All times are UTC
* There is a line of data for each recording interval (i.e. no gaps); null data are represented by -999.0

matlab
Microcat data from each download are also contained in matlab format files named catxxxx_y.mat (xxxx and y as above), with data vectors named:
catxxxx_y_cond = conductivity (mS/cm)
catxxxx_y_date = ddmmyyyy (UTC), corrected for clock drift if applicable
catxxxx_y_date_uncorrected = date as above, but no correction for clock drift
catxxxx_y_dectime = decimal time (UTC), as defined previously
catxxxx_y_dectime_uncorrected = decimal time (UTC), but no correction for clock drift
catxxxx_y_press = pressure (dbar)
catxxxx_y_sal = salinity (PSS78)
catxxxx_y_temp = temperature (^{o}C, ITS90)
catxxxx_y_time = hhmmss (UTC), corrected for clock drift if applicable
catxxxx_y_time_uncorrected = time as above, but no correction for clock drift
catxxxx_y_p = assumed constant pressure value (for AM01 microcats only)

* Null data are represented by NaN

unified data
Data from all downloads are concatenated into single files for each instrument, in ascii files named catxxxx_all.dat, and in matlab files named catxxxx_all.mat (where xxxx = microcat serial number).

additional files
* README files are included, for each download, and for concatenated files; named:
READMEcatambb_y where:
bb = borehole number
y = download number (y = "all" for concatenated files)

* Station information (lat/long/time/depth etc) are summarised in the file station.dat

5.2 CTD data

2 dbar averaged FSI CTD data from iceshelf boreholes are contained in ascii and matlab format files, as follows.

ascii

CTD data for each station are contained in 2 text files named:
fsiambbxxx.dyz_av for downcast data
fsiambbxxx.uyz_av for upcast data
and where:
bb = borehole number ("02" for AM02, "01" for AM01, "01b" for AM01-B)
xxx = CTD cast number
y = calibration status with respect to in situ borehole salinity samples:
"c" = data corrected using in situ borehole salinity samples
"x" = no useable in situ borehole salinity samples
z = calibration status with respect to ship-based calibration data:
"c" = data corrected using ship-based calibration data
"x" = no ship-based correction available
"av" indicates 2 dbar averaged data
Each file contains 2 header lines, followed by the data in column format. The columns are:
1. conductivity (mS/cm)
2. temperature (^{o}C, ITS90)
3. pressure (dbar)
4. to 6. time (UTC) (hh mm ss)
7. to 9. date (UTC) (mm dd yyyy)
10. salinity (PSS78)
11. digitiser channel number 3 (bottom contact switch)
12. digitiser channel number 4 (not used)
13. number of data points used to form 2 dbar average

* There is a line of data for each 2 dbar bin, and missing values are filled by blanks.

matlab

2 dbar averaged borehole CTD data for each season are contained in a matlab format file named fsiambb.mat, containing both downcast and upcast data, and where:
bb = borehole number ("02" for AM02, "01" for AM01, "01b" for AM01-B)
Each data matrix name begins with fsiambb_ where:
"fsi" indicates FSI CTD
bb = defined as above
followed by the parameter specification, as follows:
cond = conductivity (mS/cm)
date = ddmmyyyy (UTC)
dig3 = digitiser channel 3 (bottom contact switch)
dig4 = digitiser channel 4 (not used)
npts = number of data points used to form 2 dbar average
press = pressure (dbar)
sal = salinity (PSS78)
temp = temperature (^{o}C, ITS90)
time = hhmmss (UTC)
Data matrices concluding with "up" are upcast data; otherwise, downcast data

* There is a line of data for each 2 dbar bin; null data are represented by NaN

additional files

* README files are included for each season
* For AM01: fsiambb0xa.cc = complete data (i.e. not 2 dbar averaged), where x=station number, and cc indicates calibrated data using both in situ bottle sample correction, and ship-based correction.

6. REFERENCES

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